

# SUPPLEMENTARY INFORMATION FOR “FOURIER MEASUREMENTS ARE OPTIMAL FOR CGLMP BELL TESTS ON MAXIMALLY ENTANGLED STATES IN EVERY DIMENSION”

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ABSTRACT. This Supplementary Information contains the complete proofs of the two analytic theorems of the article (Theorem 3, the strip inequality for matrices of every size, and Theorem 4, the two-variable Bessis–Moussa–Villani formula with an explicit positive density), an outline of a second, distribution-free proof of both, precise statements and proofs of the continuum theorem and of the cell-embedding inequalities, a description of the computer-assisted certificate  $\text{CONE}_d$  in its three regimes, the proof of the rigidity theorem, the theorems on the two remaining clauses of Part B of Open Quantum Problem 27 (noise resistance and Kullback–Leibler discrimination) with their proofs, the main theorems of the Lean 4 formalisation, and the independent verifications and checks. Theorem numbers 1–4 refer to the article.

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## 1. LOGICAL STRUCTURE

Throughout,  $d \geq 2$  is the number of outcomes,  $N = 4d$ ,  $\tau = \frac{1}{M} \text{Tr}$  is the normalized trace on  $M_M(\mathbb{C})$ , and  $I_{\text{ME}}(d) = \frac{4}{d(d-1)} \sum_{j=1}^{d-1} (d-j) \sec \frac{\pi j}{2d}$ . The proof of Theorems 1 and 2 of the article has the following structure.

- (R) *Reduction* (Section 2; proved in earlier work[23]). Every projective strategy on  $\Phi_D$  gives a family of  $N$  projections on some  $\mathbb{C}^M$  whose deficit  $F - F_{\text{DKZ}}$  is a linear form  $\frac{1}{2} \langle v, \delta \rangle$  in the “pair counts”  $\delta$ .
- (T4) *Two-variable BMV formula* (Section 3): an explicit positive density. Analytic.
- (T3) *Strip inequality* (Section 4), deduced from (T4), with an exact defect formula and its equality case. Analytic.
- (C) *Continuum theorem* for projection-valued step fields on a circle (Section 5), from (T3). Analytic.
- (E) *Cell inequalities*  $\langle u^\ell, \delta \rangle \leq 0$  for every cell vector  $\ell$  (Section 6), from (C). Analytic.
- (K) *CONE<sub>d</sub>*:  $v$  is a non-negative combination of cell vectors  $u^\ell$  and trivial directions (Section 7). Computer-assisted (interval arithmetic) for every  $d$ .
- (A) (R)+(E)+(K) give Theorem 1 (Section 6).
- (B) The equality case of the chain gives Theorem 2 (Section 8).

Nothing else is used. In particular the exact sum-of-squares certificates of earlier work for  $3 \leq d \leq 20$  [24] are not used; they give an independent proof of Theorems 1 and 2 for those  $d$ .

Every item of this list, except (K) for  $d \geq 21$ , has been formalised in Lean 4 (Section 10). For  $2 \leq d \leq 20$  the formal proof of Theorems 1 and 2 is complete, including (K), whose certificates are re-evaluated by the Lean kernel. For  $d \geq 21$  it is complete except for the single input (K), which is established by the interval-arithmetic certificates of Section 7, outside Lean. Theorems 3 and 4 are formalised in full. Thus Theorems 1 and 2 are proved for every  $d$ , with a computer-assisted step (interval arithmetic) for  $d \geq 21$ , and fully formalised for  $d \leq 20$ .

Section 9 treats the two remaining clauses of Part B of Problem 27. It uses Theorems 1 and 2 only for the CGLMP reading of the noise clause and for  $d = 3$ ; the other results there are independent of the chain above.

## 2. THE CGLMP EXPRESSION, THE CLOCK MODEL AND THE CIRCLE CONFIGURATION

**2.1. CGLMP expression.** With outcomes  $A_1, A_2, B_1, B_2 \in \mathbb{Z}_d$ , the CGLMP expression[7] is

$$I_d = \sum_{k=0}^{\lfloor d/2 \rfloor - 1} \left( 1 - \frac{2k}{d-1} \right) \left( [P(A_1 = B_1 + k) + P(B_1 = A_2 + k + 1) + P(A_2 = B_2 + k) + P(B_2 = A_1 + k)] - [P(A_1 = B_1 - k - 1) + P(B_1 = A_2 - k) + P(A_2 = B_2 - k - 1) + P(B_2 = A_1 - k - 1)] \right),$$

where  $P(A_x = B_y + k)$  is the probability that  $A_x - B_y \equiv k \pmod{d}$ . Let  $m(t) \in \{0, \dots, d-1\}$  be the residue of  $t$  modulo  $d$ , relabel  $X_1 = A_2$ ,  $Y_1 = B_2$ ,  $X_2 = A_1$ ,  $Y_2 = B_1$ , and put  $S = \mathbb{E} m(X_1 - Y_1) + \mathbb{E} m(Y_1 - X_2) + \mathbb{E} m(X_2 - Y_2) + \mathbb{E} m(Y_2 - X_1 - 1)$ . Then  $I_d = 4 - \frac{2}{d-1} S$  for every behaviour (ref. [23], Lemma 2.1), and the local bound  $I_d \leq 2$  is  $S \geq d-1$ .

**2.2. Maximally entangled strategies.** For projective measurements  $\{P_a^x\}, \{Q_b^y\}$  on  $\mathbb{C}^D \otimes \mathbb{C}^D$  and  $\Phi_D = D^{-1/2} \sum_j |jj\rangle$ , one has  $P(a, b|x, y) = \tau(P_a^x(Q_b^y)^\top)$  with  $\tau = \frac{1}{D} \text{Tr}$ . With  $w = e^{2\pi i/d}$ ,  $U_x = \sum_a w^a P_a^x$ ,  $U'_y = \sum_b w^b Q_b^y$  and  $R_1 = U_2$ ,  $R_2 = (U'_2)^\top$ ,  $R_3 = U_1$ ,  $R_4 = (U'_1)^\top$  (so  $R_i^d = \mathbf{1}$ ),

the expansion  $m(t) = \frac{d-1}{2} + \sum_{n=1}^{d-1} c_n w^{nt}$ ,  $c_n = -1/(1 - w^{-n})$ , gives

$$(2.1) \quad S = 2(d-1) + \sum_{n=1}^{d-1} c_n \sum_{i=0}^3 \tau(L_n^{(i)})$$

with the links  $L_n^{(0)} = R_1^n R_2^{-n}$ ,  $L_n^{(1)} = R_2^n R_3^{-n}$ ,  $L_n^{(2)} = R_3^n R_4^{-n}$  and  $L_n^{(3)} = w^{-n} R_4^n R_1^{-n}$ . Conversely every quadruple of unitaries with  $R_i^d = \mathbf{1}$  comes from a projective strategy on  $\Phi_D$ .

**2.3. The clock model.** Let  $z = e^{2\pi i/N}$ . For  $V_0, \dots, V_{d-1} \in U(M)$  with  $V_k^4 = \mathbf{1}$  define the covariant strategy  $W = \sum_k |k\rangle\langle k| \otimes z^k V_k$ ,  $R_1^V = X \otimes \mathbf{1}_M$ ,  $R_{i+1}^V = W R_i^V W^\dagger$ , with  $X$  the cyclic shift on  $\mathbb{C}^d$ .

**Theorem 2.1** (reduction; ref. [23], Theorem 2.3). (1)  $I_d(R^V) = 4F(V)/(d(d-1))$  with  $F(V) = \sum_{j < k} \text{Re}[h_{k-j} \text{tr}(V_j V_k^\dagger)]$ ,  $h_m = \sec \psi_m - i \csc \psi_m$ ,  $\psi_m = \pi m/(2d)$ . (2) Every projective strategy  $R$  on  $\Phi_D$  is unitarily equivalent to a direct summand of some  $R^V$  (one can take  $M = 4D$ ) with  $I_d(R) = I_d(R^V)$ . The family  $V$  is unique up to simultaneous conjugation. (3) The DKZ strategy is, up to the local unitary  $G \otimes \bar{G}$  with  $G = W_0^{-2}$ ,  $W_0 = \text{diag}(z^k)$ , the covariant strategy with  $M = 1$ ,  $V \equiv 1$ ; hence  $I_{\text{ME}}(d) = 4F_{\text{DKZ}}/(d(d-1))$  with  $F_{\text{DKZ}} = \sum_{m=1}^{d-1} (d-m) \sec \psi_m$ .

The family  $V$  in (2) is obtained by direct-summing the images of  $R$  under the cyclic group of order  $4d$  generated by  $\rho(R_1, R_2, R_3, R_4) = (R_2, R_3, R_4, wR_1)$  and applying the Stone–von Neumann theorem for  $\mathbb{Z}_d$  to the shift of the summands; see ref. [23], Appendix A.

**2.4. The circle configuration.** Let  $E_k = V_k^\dagger$ . Write  $x \in \mathbb{Z}_N$  uniquely as  $x = k - da$  ( $0 \leq k \leq d-1$ ,  $a \in \mathbb{Z}_4$ ) and let  $Q_x$  be the spectral projection of  $E_k$  for the eigenvalue  $i^a$ . For each  $k$ ,  $\{Q_{k-da}\}_{a \in \mathbb{Z}_4}$  is a projective measurement and  $E_k = \sum_a i^a Q_{k-da}$ . We call such a family of  $N$  projections a *27B configuration*. Put  $A_x = Q_x$ ,  $B_x = Q_{x+d}$  (so  $A_x B_x = 0$  and  $\sum_x A_x = \sum_x B_x = d\mathbf{1}$ ),  $\kappa(u) = \cot(\pi u/N)$  for  $u \neq 0$ ,  $\kappa(0) = 0$ , and

$$\langle A, B \rangle_\tau = \sum_{x, y \in \mathbb{Z}_N} \kappa(x-y) \tau(A_x B_y), \quad T(m) = \sum_y \tau(A_{y+m} B_y), \quad N(m) = T(m) - T(-m).$$

**Lemma 2.2** (linear form).  $F(V) = \frac{1}{2} \langle A, B \rangle_\tau - \frac{d}{2}$ ,  $N(d) = d$ ,  $N(2d-m) = N(m)$ , and

$$F(V) - F_{\text{DKZ}} = \frac{1}{2} \langle v, \delta \rangle, \quad v_m = 2 \csc \frac{\pi m}{2d}, \quad \delta_m = N(m) - m \quad (1 \leq m \leq d-1).$$

Moreover, for  $1 \leq s \leq d-1$ ,  $\delta_s + \delta_{d-s} \leq 0$ .

*Proof.* The first identity is the projection form of the cotangent representation of ref. [23] (Lemma 3.1 there, and its operator version in Section 6); its proof holds verbatim for non-commuting  $V_k$ . For the others, put  $C(n) = \sum_u \tau(Q_u Q_{u+n})$ . Then

$$T(m) = C(m-d), \quad N(m) = C(m-d) - C(m+d).$$

Since  $Q_u Q_{u+2d} = 0$  (two outcomes of one measurement) and  $\sum_u \tau(Q_u) = d$ ,  $N(d) = C(0) - C(2d) = d$ , and, by  $C(-n) = C(n)$  and  $N$ -periodicity,  $N(2d-m) = C(d-m) - C(3d-m) = C(m-d) - C(m+d) = N(m)$ . As  $\kappa$  is odd and  $\kappa(2d) = 0$ ,  $\langle A, B \rangle_\tau = \sum_{m=1}^{2d-1} \kappa(m) N(m) = d\kappa(d) + \sum_{m=1}^{d-1} (\kappa(m) + \kappa(2d-m)) N(m)$ , and  $\kappa(m) + \kappa(2d-m) = \cot \frac{\pi m}{4d} + \tan \frac{\pi m}{4d} = 2 \csc \frac{\pi m}{2d}$ . For  $V \equiv 1$  the configuration is  $A = [0, d)$ ,  $B = [-d, 0)$ , with  $N(m) = m$ . For the last claim, let  $\pi_{jk}(t) = \sum_b \tau(Q_{j-db} Q_{k-d(b+t)})$  for sites  $j < k$ , a probability vector on  $\mathbb{Z}_4$ . Splitting the sum defining  $N(m)$  according to whether the site index wraps around gives  $N(m) = \sum_{k-j=m} [\pi_{jk}(1) - \pi_{jk}(3)] + \sum_{k-j=d-m} [\pi_{jk}(0) - \pi_{jk}(2)]$ , so  $N(s) + N(d-s) - d$  is a sum over the  $d$  pairs at distance  $s$  or  $d-s$  of  $\pi(0) + \pi(1) - \pi(2) - \pi(3) - 1 \leq 0$ .  $\square$

Hence Theorem 1 is equivalent to  $\langle v, \delta \rangle \leq 0$  for every 27B configuration on every  $\mathbb{C}^M$ .

## 3. THEOREM 4: THE TWO-VARIABLE BMV FORMULA WITH AN EXPLICIT DENSITY

Throughout this section and the next,  $B$  is an orthogonal projection of rank  $r$  on  $\mathbb{C}^M$ ,  $P = \mathbf{1} - B$ ,  $g$  is Hermitian,  $g_d = BgB + PgP$ , and  $\lambda_{\min} \leq \lambda_{\max}$  are the extreme eigenvalues of  $g$ .  $\text{Tr}_P$  and  $\text{Tr}_B$  denote traces over the ranges of  $P$  and  $B$  of the compressions. For  $(a, t) \in \mathbb{C}^2$ ,

$$\mathcal{D}(a, t) = \text{Tr} e^{ag - tP} - e^{-t} \text{Tr}_P e^{aPgP} - \text{Tr}_B e^{aBgB}.$$

For  $s \in \mathbb{R}$  and  $0 < \tau < 1$  let  $p_{s,\tau}(x) = \det(g - s - x(P - \tau))$  and  $q_{s,\tau}(x) = \det(g_d - s - x(P - \tau))$ , let  $x_1(s, \tau), \dots, x_M(s, \tau)$  be the roots of  $p_{s,\tau}$  (the eigenvalues of  $(P - \tau)^{-1}(g - s)$ ), and

$$F(s, \tau) = \frac{1}{2\pi} \sum_{i=1}^M |\text{Im} x_i(s, \tau)| \quad (0 < \tau < 1), \quad F(s, \tau) = 0 \text{ otherwise.}$$

**Theorem 3.1** (Theorem 4 of the article, except the defect formula).  *$F$  is non-negative and continuous on  $\mathbb{R} \times (0, 1)$ , vanishes unless  $\lambda_{\min} < s < \lambda_{\max}$ , satisfies  $F(s, \tau) \leq \frac{M}{2\pi} \|g - s\| / \sqrt{\tau(1 - \tau)}$  (so  $F \in L^1(\mathbb{R}^2)$  with compact support), and for all  $(a, t) \in \mathbb{C}^2$*

$$(3.1) \quad \mathcal{D}(a, t) = a^2 \int_0^1 \int_{\mathbb{R}} e^{as - t\tau} F(s, \tau) ds d\tau.$$

**Lemma 3.2** (pencil facts). *Let  $0 < \tau < 1$  and  $s \in \mathbb{R}$ . (a) Every root of  $p_{s,\tau}$  satisfies  $|x| \leq \|g - s\| / \min(\tau, 1 - \tau)$ , and so does every root of  $q_{s,\tau}$ . (b) Every non-real root of  $p_{s,\tau}$  satisfies  $|x| \leq \|g - s\| / \sqrt{\tau(1 - \tau)}$ . (c) If  $s \leq \lambda_{\min}$  or  $s \geq \lambda_{\max}$ , all roots of  $p_{s,\tau}$  are real. (d) All roots of  $q_{s,\tau}$  are real. (e)  $p_{s,\tau}$  and  $q_{s,\tau}$  have degree  $M$ , the same leading coefficient  $\det(\tau - P) \neq 0$  and the same sum of roots. (f)  $F$  is continuous on  $\mathbb{R} \times (0, 1)$ .*

*Proof.* A root  $x$  comes with a unit vector  $\psi$  such that  $(g - s)\psi = x(P - \tau)\psi$ . (a)  $P - \tau$  has eigenvalues  $1 - \tau$  and  $-\tau$ , so  $|x| \min(\tau, 1 - \tau) \leq |x| \|(P - \tau)\psi\| = \|(g - s)\psi\| \leq \|g - s\|$ ; likewise for  $q$  with  $\|g_d - s\| \leq \|g - s\|$ . (b)  $\langle \psi, (g - s)\psi \rangle = x \langle \psi, (P - \tau)\psi \rangle$  with both inner products real; if  $\text{Im} x \neq 0$  then  $\langle \psi, (P - \tau)\psi \rangle = 0$ , i.e.  $\|P\psi\|^2 = \tau$ ,  $\|B\psi\|^2 = 1 - \tau$ , hence  $\|(P - \tau)\psi\|^2 = (1 - \tau)^2\tau + \tau^2(1 - \tau) = \tau(1 - \tau)$  and the argument of (a) applies. (c) If  $g - s$  is semidefinite and  $\text{Im} x \neq 0$ , then by (b)  $\langle \psi, (g - s)\psi \rangle = 0$ , so  $(g - s)\psi = 0$ , so  $x(P - \tau)\psi = 0$  with  $x \neq 0$  and  $P - \tau$  invertible, a contradiction. (d)  $g_d - s - x(P - \tau)$  is block diagonal; its determinant vanishes iff  $x = (s - \mu)/\tau$  with  $\mu \in \text{spec}(BgB|_{\text{ran} B})$  or  $x = (\mu - s)/(1 - \tau)$  with  $\mu \in \text{spec}(PgP|_{\text{ran} P})$ . (e)  $p_{s,\tau}(x) = \det(\tau - P) \det(x - (P - \tau)^{-1}(g - s))$ , so the roots sum to  $\text{Tr}[(P - \tau)^{-1}(g - s)]$ , and likewise for  $q$ ; since  $(P - \tau)^{-1} = P/(1 - \tau) - B/\tau$  commutes with  $B$ ,  $\text{Tr}[(P - \tau)^{-1}(g - g_d)] = \text{Tr}[(P - \tau)^{-1}(BgP + PgB)] = 0$ . (f) The coefficients of  $p_{s,\tau}$  are polynomials in  $(s, \tau)$  and the leading coefficient does not vanish on  $(0, 1)$ , so the roots depend continuously on  $(s, \tau)$  as an unordered  $M$ -tuple, and  $\sum_i |\text{Im} x_i|$  is a continuous symmetric function of it.  $\square$

**Lemma 3.3** (Jensen-type identity). *Let  $p(x) = c \prod_{i=1}^M (x - x_i)$  and  $q(x) = c \prod_{i=1}^M (x - y_i)$  with  $c \neq 0$ , all  $y_i$  real, and  $\sum_i x_i = \sum_i y_i$ . Then  $\log |p/q| \in L^1(\mathbb{R})$  and  $\int_{\mathbb{R}} \log |p(\xi)/q(\xi)| d\xi = \pi \sum_i |\text{Im} x_i|$ .*

*Proof.* Write  $x_i = \alpha_i + i\beta_i$ . Then  $\log |p/q| = A + C$  with  $A(\xi) = \sum_i \frac{1}{2} \log(1 + \beta_i^2/(\xi - \alpha_i)^2) \geq 0$  and  $C(\xi) = \sum_i [\log |\xi - \alpha_i| - \log |\xi - y_i|]$ . Each summand of  $A$  is integrable with  $\int_{\mathbb{R}} \frac{1}{2} \log(1 + b^2/u^2) du = \pi|b|$  (substitute  $u = |b|v$  and use the antiderivative  $v \log(1 + v^{-2}) + 2 \arctan v$ ).  $C$  has only logarithmic singularities, and  $C(\xi) = O(\xi^{-2})$  as  $|\xi| \rightarrow \infty$  because  $\sum_i \alpha_i = \sum_i y_i$ ; so  $C \in L^1$ . For  $L > \max_i(|\alpha_i|, |y_i|)$ ,  $\int_{-L}^L \log |\xi - a| d\xi = m_L(a) := (L - a) \log(L - a) + (L + a) \log(L + a) - 2L$ , which is even and smooth in  $a$  with  $m_L''(a) = 2L/(L^2 - a^2)$ , so  $m_L(a) = m_L(0) + a^2/L + O(a^4/L^3)$ . Hence  $\int_{-L}^L C = \sum_i (\alpha_i^2 - y_i^2)/L + O(L^{-3}) \rightarrow 0$ , and  $\int_{\mathbb{R}} C = 0$ .  $\square$

**Lemma 3.4** (real slices). *For  $\xi \in \mathbb{R}$  let  $\lambda_j(\xi)$  and  $\lambda_j^0(\xi)$  ( $j = 1, \dots, M$ ) be the eigenvalues of  $g - \xi P$  and of  $g_d - \xi P$ , and put  $U_\xi(w) = \sum_j [(w - \lambda_j(\xi))_+ - (w - \lambda_j^0(\xi))_+]$  and  $N_\xi(w) =$*

$\sum_j [\mathbf{1}_{w > \lambda_j(\xi)} - \mathbf{1}_{w > \lambda_j^0(\xi)}]$ . Then (a)  $U_\xi$  is continuous, piecewise linear and compactly supported, with  $U'_\xi = N_\xi$  almost everywhere; (b)  $\int_{\mathbb{R}} e^{aw} U_\xi(w) dw = \mathcal{D}(a, a\xi)/a^2$  for every complex  $a \neq 0$ ; (c) the Hilbert transform  $(Hf)(w) = \frac{1}{\pi} \text{p.v.} \int f(u)/(w-u) du$  of  $N_\xi$  is  $(HN_\xi)(w) = \frac{1}{\pi} \log |\det(g - \xi P - w)/\det(g_d - \xi P - w)|$  for  $w$  outside both spectra.

*Proof.* (a) Below all eigenvalues  $U_\xi = 0$ ; above all of them  $U_\xi = \text{Tr}(g_d - \xi P) - \text{Tr}(g - \xi P) = 0$ . (b)  $U''_\xi = \sum_j (\delta_{\lambda_j} - \delta_{\lambda_j^0})$  as distributions and  $U_\xi$  has compact support, so two integrations by parts give  $a^{-2} \sum_j (e^{a\lambda_j} - e^{a\lambda_j^0}) = a^{-2} [\text{Tr} e^{a(g-\xi P)} - \text{Tr}_B e^{aBgB} - e^{-a\xi} \text{Tr}_P e^{aPgP}]$ . (c) Pair  $\lambda_j$  with  $\lambda_j^0$ ; each difference of indicators is  $\pm$  the indicator of the interval between them, and  $\frac{1}{\pi} \int_\alpha^\beta du/(w-u) = \frac{1}{\pi} \log |(w-\alpha)/(w-\beta)|$ .  $\square$

**Lemma 3.5** (limits). *For Hermitian  $A$ ,*

$$\lim_{t \rightarrow +\infty} \text{Tr} e^{A-tP} = \text{Tr}_B e^{BAB}, \quad \lim_{t \rightarrow -\infty} e^t \text{Tr} e^{A-tP} = \text{Tr}_P e^{PAP}.$$

*Proof.* Shifting  $A$  by a constant multiplies both sides by the same factor, so let  $A \geq \mathbf{1}$ . Let  $\mu_1 \geq \dots \geq \mu_M$  be the eigenvalues of  $H_t = A - tP$  and  $\beta_1 \geq \dots \geq \beta_r$  those of  $BAB|_{\text{ran} B}$ . Min-max over subspaces of  $\text{ran} B$  gives  $\mu_k \geq \beta_k$  ( $k \leq r$ ). For  $\psi = b + p$  ( $b \in \text{ran} B$ ,  $p \in \text{ran} P$ ) and  $t > \|A\|$ ,  $\langle \psi, H_t \psi \rangle \leq \langle b, Ab \rangle + 2\|A\|\|b\|\|p\| - (t - \|A\|)\|p\|^2 \leq \langle b, (A + \epsilon_t)b \rangle$  with  $\epsilon_t = \|A\|^2/(t - \|A\|)$ , so  $\mu_k \leq \beta_k + \epsilon_t$  for  $k \leq r$ , and  $\mu_{r+1} \leq \|A\| - t$  (min-max with the codimension- $r$  subspace  $\text{ran} P$ ). Hence  $\text{Tr} e^{H_t} \rightarrow \sum_k e^{\beta_k}$ . The second limit is the first with  $B$  and  $P$  exchanged, since  $e^t \text{Tr} e^{A-tP} = \text{Tr} e^{A+tB}$ .  $\square$

*Proof of Theorem 3.1.* The properties of  $F$  are Lemma 3.2. *Step 0.*  $\mathcal{D}$  is entire on  $\mathbb{C}^2$ . Since  $e^{-tP} = B + e^{-t}P$ ,  $\mathcal{D}(0, t) = 0$  and  $\partial_a \mathcal{D}(0, t) = \text{Tr}(ge^{-tP}) - e^{-t} \text{Tr}(PgP) - \text{Tr}(BgB) = 0$ . Hence  $G(a, t) := \mathcal{D}(a, t)/a^2 = \int_0^1 (1-u) \partial_a^2 \mathcal{D}(ua, t) du$  is entire.

*Step 1 (a distribution).* Put  $\hat{G}(\zeta) = G(-i\zeta_1, i\zeta_2)$ , so that  $e^{-i(\zeta_1 s + \zeta_2 \tau)} = e^{as - t\tau}$  for  $a = -i\zeta_1$ ,  $t = i\zeta_2$ . Let  $K = [\lambda_{\min}, \lambda_{\max}] \times [0, 1]$  and  $H_K(\eta) = \sup_{(s, \tau) \in K} (s\eta_1 + \tau\eta_2)$ . For complex  $a, t$  the Hermitian part of  $ag - tP$  is  $\eta_1 g + \eta_2 P$  with  $\eta = \text{Im} \zeta$ , so  $\|e^{ag-tP}\| \leq e^{\lambda_{\max}(\eta_1 g + \eta_2 P)} \leq e^{H_K(\eta)}$  (the pair  $(\langle \psi, g\psi \rangle, \langle \psi, P\psi \rangle)$  lies in  $K$  for unit  $\psi$ ). The two subtracted terms are bounded in the same way, so  $|\mathcal{D}| \leq 2Me^{H_K(\eta)}$ . For  $|\zeta_1| \geq 1$  this bounds  $\hat{G}$ ; for  $|\zeta_1| < 1$  the maximum principle on  $|\zeta_1| \leq 2$  gives  $|\hat{G}(\zeta)| \leq \frac{M}{2} e^{H_K(\eta) + 3c_K}$  with  $c_K = \max(|\lambda_{\min}|, |\lambda_{\max}|)$ . By the Paley–Wiener–Schwartz theorem (ref. [14], Theorem 7.3.1) there is a distribution  $T$  supported in  $K$  with

$$(3.2) \quad T(e^{as-t\tau}) = G(a, t) \quad \text{for all } (a, t) \in \mathbb{C}^2,$$

and  $\hat{G}$  is bounded on  $\mathbb{R}^2$ .

*Step 2 ( $T = F$  on the open strip).* Let  $\varphi \in C_c^\infty(\mathbb{R}^2)$  with Fourier transform  $\hat{\varphi}$ . Since  $T$  has compact support and bounded transform,  $T(\varphi) = (2\pi)^{-2} \int_{\mathbb{R}^2} \hat{G}(k) \hat{\varphi}(-k) dk$ . Change variables  $k = (\sigma, -\sigma\xi)$  (Jacobian  $|\sigma|$ ) and use Fubini:  $T(\varphi) = (2\pi)^{-2} \int d\xi \int d\sigma |\sigma| \hat{G}(\sigma, -\sigma\xi) \hat{\varphi}(-\sigma, \sigma\xi)$ . By Lemma 3.4(b),  $\hat{G}(\sigma, -\sigma\xi) = \hat{U}_\xi(\sigma)$ , the Fourier transform of  $U_\xi$ ; and  $\hat{\varphi}(-\sigma, \sigma\xi) = \hat{\psi}_\xi(-\sigma)$  with  $\psi_\xi(w) = \int \varphi(w + \xi\tau, \tau) d\tau$ . By Lemma 3.4(a),  $|\sigma| \hat{U}_\xi = (-i \text{sgn } \sigma) \hat{N}_\xi = \widehat{HN_\xi}$ , so by Parseval the inner integral is  $2\pi \iint (HN_\xi)(s - \xi\tau) \varphi(s, \tau) ds d\tau$ , and by Lemma 3.4(c) with  $g - \xi P - (s - \xi\tau) = g - s - \xi(P - \tau)$ ,

$$T(\varphi) = \frac{1}{2\pi^2} \int_{\mathbb{R}} d\xi \iint \varphi(s, \tau) \log \left| \frac{p_{s, \tau}(\xi)}{q_{s, \tau}(\xi)} \right| ds d\tau.$$

If  $\text{supp } \varphi \subset [-S, S] \times [\delta, 1 - \delta]$ , all roots of  $p_{s, \tau}, q_{s, \tau}$  lie in  $|x| \leq R = (\|g\| + S)/\delta$  there (Lemma 3.2(a)); the integrand is uniformly integrable for  $|\xi| \leq 2R$  (logarithmic singularities), and for  $|\xi| > 2R$  it is bounded by  $2MR^2/\xi^2$ , using  $|\log |1 - z| + \text{Re } z| \leq |z|^2$  for  $|z| \leq \frac{1}{2}$  and the equal root sums. So Fubini applies, and Lemma 3.3 (with Lemma 3.2(d),(e)) gives  $T(\varphi) = \iint \varphi F$ .

*Step 3 (no mass on the edges).*  $T - F$  has compact support contained in the two segments  $[\lambda_{\min}, \lambda_{\max}] \times \{0\}$  and  $\times \{1\}$ . By the structure theorem for distributions supported in a hyperplane (ref. [14], Theorem 2.3.5),  $T - F = \sum_{k \leq n} [\alpha_k \otimes \delta^{(k)}(\tau) + \beta_k \otimes \delta^{(k)}(\tau - 1)]$  with compactly supported distributions  $\alpha_k, \beta_k$ , and therefore

$$G(a, t) - \iint e^{as-t\tau} F ds d\tau = \sum_k t^k [A_k(a) + e^{-t} B_k(a)], \quad A_k(a) = \alpha_k(e^{as}), \quad B_k(a) = \beta_k(e^{as}).$$

Fix real  $a \neq 0$ . As  $t \rightarrow +\infty$ ,  $G(a, t) \rightarrow 0$  by Lemma 3.5 and the integral tends to 0 by dominated convergence, so the polynomial  $\sum_k A_k(a) t^k$  tends to 0 and all  $A_k(a) = 0$ . As  $t \rightarrow -\infty$ , multiplying by  $e^t$  and using Lemma 3.5 again gives  $B_k(a) = 0$ . The  $A_k, B_k$  are entire and vanish on  $\mathbb{R} \setminus \{0\}$ , hence identically, so  $\alpha_k = \beta_k = 0$  and  $T = F$ . Equation (3.2) is equation (3.1).  $\square$

*Remark 3.6 (sum rule).* The second-order Duhamel formula gives  $G(0, t) = \|BgP\|_F^2 (1 - e^{-t})/t = \|BgP\|_F^2 \int_0^1 e^{-t\tau} d\tau$ . By Theorem 3.1,  $\int_{\mathbb{R}} F(s, \tau) ds = \|BgP\|_F^2$  for every  $\tau \in (0, 1)$ . This is the horizontal Radon projection, which the proof does not use; it is an independent check (Fig. 4d of the article).

*Remark 3.7 ( $M = 2$ ).* For  $B = \text{diag}(1, 0)$  and  $g = \begin{pmatrix} \alpha & c \\ c & \beta \end{pmatrix}$  the discriminant of  $p_{s,\tau}$  is  $(m(\tau) - s)^2 - 4\tau(1-\tau)|c|^2$  with  $m(\tau) = \tau\beta + (1-\tau)\alpha$ , so  $F(s, \tau) = \sqrt{4\tau(1-\tau)|c|^2 - (s - m(\tau))^2} / (2\pi\tau(1-\tau))$ : every slice is a semicircle law of mass  $|c|^2$  centred on the segment from  $\alpha$  ( $\tau = 0$ ) to  $\beta$  ( $\tau = 1$ ).

*Remark 3.8 (structure).*  $(P - \tau)^{-1}(g - s)$  is self-adjoint for the indefinite form  $[u, u'] = \langle u, (P - \tau)u' \rangle$ , which has  $M - r$  positive and  $r$  negative squares. Hence at most  $\min(r, M - r)$  conjugate pairs of non-real roots occur, and for rank  $B = 1$ ,  $F = \frac{1}{\pi} \text{Im } x_+$  for the single root in the upper half-plane (if any).

**Corollary 3.9** (Stahl's theorem for projections with an explicit density). *For Hermitian  $A$  and an orthogonal projection  $P$ ,  $t \mapsto \text{Tr } e^{A-tP}$  is the Laplace transform of the positive measure  $\text{Tr}_B(e^{BAB})\delta_0 + \text{Tr}_P(e^{PAP})\delta_1 + (\int_{\mathbb{R}} e^s F_A(s, \tau) ds) d\tau$  on  $[0, 1]$ , where  $F_A$  is the density of Theorem 3.1 for  $g = A$ .*

*Proof.* Take  $a = 1$ ,  $g = A$  in equation (3.1).  $\square$

*Remark 3.10 (prior work).* Stahl proved that  $t \mapsto \text{Tr } e^{A-tB}$  is the Laplace transform of a positive measure for all Hermitian  $A$  and  $B$  [27, 9], settling the conjecture of Bessis, Moussa and Villani [3, 20]. Heinävaara [13] constructed tracial joint spectral measures by a two-dimensional Fourier inversion whose density is given by the imaginary parts of eigenvalues of an auxiliary pencil, and derived Stahl's theorem from it. Theorem 3.1 is a variant of this mechanism, adapted to a projection, to the pinched subtraction and to the division by  $a^2$ ; the same proof gives the analogous formula for general Hermitian  $B$ , with atoms at all eigenvalues of  $B$ . We found the proof independently and learned of ref. [13] afterwards. What is specific to the present setting is the positivity of the pinched,  $a^2$ -divided object and its consequence, the strip inequality.

**An elementary route.** Steps 1–3 of the proof of Theorem 3.1 use the Paley–Wiener–Schwartz theorem, Fourier inversion and the structure of distributions supported on a line. They can be replaced by an argument without distributions, based on the following identity.

**Proposition 3.11** (Radon identity). *Let  $H$  be Hermitian on  $\mathbb{C}^M$  and, for  $0 < \tau < 1$ , let  $y_1(\tau), \dots, y_M(\tau)$  be the eigenvalues of  $(P - \tau)^{-1}H$ . Then  $\tau \mapsto \sum_i |\text{Im } y_i(\tau)|$  is continuous on  $(0, 1)$ , bounded by  $M\|H\|/\sqrt{\tau(1-\tau)}$ , and*

$$\frac{1}{2\pi} \int_0^1 \sum_{i=1}^M |\text{Im } y_i(\tau)| d\tau = \text{Tr } H_+ - \text{Tr}(BHB + PHP)_+.$$

*Outline of the proof.* Replace  $H$  by  $H + c$  with  $\operatorname{Im} c > 0$ . An eigenvector  $\psi$  of a root  $y$  gives  $\operatorname{Im} c = \langle \psi, (P - \tau)\psi \rangle \operatorname{Im} y$ , so no root is real, and a homotopy  $H \rightarrow sH$  shows that exactly  $\operatorname{rank} P$  roots lie in the upper half-plane. Since  $\det(H + c - y(P - \tau))$  depends on  $(\tau, c)$  only through  $(y, c + y\tau)$ , a simple root satisfies the complex inviscid Burgers equation  $\partial_\tau y = y \partial_c y$ ; summed over the roots in the upper half-plane with a Cauchy integral, which is valid also at repeated roots, this gives the conservation law  $\partial_\tau \sum \operatorname{Log} y = \partial_c \sum y$ . Integrating over  $\tau \in (0, 1)$ , the end  $\tau \rightarrow 1$  contributes nothing after the pinched terms are subtracted, and the end  $\tau \rightarrow 0$  gives, through a Schur complement,  $\log \det(H + c) - \log \det(H_d + c)$  with  $H_d = BHB + PHP$ . Hence the  $\tau$ -integral  $\Psi(c)$  of the sum of the roots in the upper half-plane, minus its pinched counterpart, equals  $\mathcal{E}(c) = \operatorname{Tr}[(H_d + c)\operatorname{Log}(H_d + c)] - \operatorname{Tr}[(H + c)\operatorname{Log}(H + c)]$  up to a term  $-2\pi i k c + \kappa_0$  with  $k \in \mathbb{Z}$  and  $\kappa_0 \in \mathbb{C}$ . As  $c$  tends to a real  $c_0$ , dominated convergence (the integrand is bounded by  $M(\|H\| + |c|)(2 + (\tau(1 - \tau))^{-1/2})$ : the argument of Lemma 3.2(b) gives  $|y| \leq (\|H\| + |c|)/\sqrt{\tau(1 - \tau)}$  for the roots in the upper half-plane when  $\tau \leq \frac{1}{2}$  and for those in the lower half-plane when  $\tau \geq \frac{1}{2}$ , and the sum of all roots equals that of the pinched pencil) turns the imaginary parts into  $\pi$  times the two sides of the identity for  $H + c_0$ , up to  $-2\pi k c_0 + \operatorname{Im} \kappa_0$ . For  $c_0 > \|H\|$  both sides vanish, because all roots are then real; hence  $k = 0$ ,  $\operatorname{Im} \kappa_0 = 0$ , and the identity holds. The complete proof is in the repository[25].

*Consequences.* On the line  $s = w + \xi\tau$  the roots of  $p_{s,\tau}$  are  $\xi + y_i$ , where the  $y_i$  are those of Proposition 3.11 for  $H = g - \xi P - w$ , and  $BHB + PHP = g_d - \xi P - w$ . So the identity says that the integral of  $F$  along this line equals  $U_\xi(w)$  (Lemma 3.4): the Radon projections of  $F$  are the real slices. By Lemma 3.4(b) and Tonelli's theorem, equation (3.1) holds for all real  $a \neq 0$  and real  $t$ ; both sides are entire on  $\mathbb{C}^2$ , so it holds everywhere by the identity theorem, applied in  $a$  and then in  $t$ . In the proof of Theorem 4.3, step (b) then needs only a one-variable uniqueness lemma: a continuous  $Z$  with  $|Z(w)| \leq C e^{-c|w|}$  whose two-sided Laplace transform vanishes on an interval is zero (analytic continuation to the imaginary axis and a Gaussian approximate identity). This route, which has been checked by an independent verification (Section 11), is the one followed by the Lean formalisation (Section 10); there the one-variable uniqueness step uses the Fourier inversion theorem of Mathlib. The same argument for an arbitrary Hermitian matrix in place of the projection  $P$  gives a proof of Stahl's theorem[27] without distribution theory or Fourier analysis; we make no priority claim about this by-product.

#### 4. THEOREM 3: THE STRIP INEQUALITY

Let  $S = \{0 \leq \operatorname{Re} \nu \leq 1\}$ . For  $\lambda \in \mathbb{R}$  and  $x + iy$  with  $0 < x < 1$  let  $h_\lambda(x + iy) = -\frac{1}{\pi^2} \operatorname{Im} \operatorname{Li}_2(e^{\pi(y-\lambda)} e^{-i\pi x})$ ; the argument  $-\pi x \in (-\pi, 0)$  never meets the cut  $[1, \infty)$  of  $\operatorname{Li}_2$ , so  $h_\lambda$  is harmonic. For  $0 < X < 1$  let  $K_X(u) = \sin(\pi X)/(2(\cosh \pi u - \cos \pi X))$ .

**Lemma 4.1** (strip Poisson kernel).  $K_X > 0$ ,  $K_X(u) \leq C_X e^{-\pi|u|}$ ,  $K_X(u) = \sum_{n \geq 1} \sin(n\pi X) e^{-n\pi|u|}$  for  $u \neq 0$ , and for  $|\operatorname{Re} a| < \pi$ ,  $\int_{\mathbb{R}} e^{-au} K_X(u) du = \sin(a(1 - X))/\sin a$ .

*Proof.* The series follows from  $\sum_n q^n \sin n\theta = q \sin \theta / (1 - 2q \cos \theta + q^2)$  with  $q = e^{-\pi|u|}$ ,  $\theta = \pi X$ . For the transform take  $a = i\kappa$ ,  $\kappa \neq 0$  real, and integrate  $f(u) = e^{-i\kappa u} / (\cosh \pi u - \cos \pi X)$  over the boundary of  $[-L, L] \times [0, 2]$ . Since  $f(u + 2i) = e^{2\kappa} f(u)$ , the contour integral is  $(1 - e^{2\kappa}) \int_{\mathbb{R}} f$ ; the poles inside are  $u = iX$  and  $u = i(2 - X)$  with residues  $e^{\kappa X} / (i\pi \sin \pi X)$  and  $-e^{\kappa(2-X)} / (i\pi \sin \pi X)$ . Hence  $\int e^{-i\kappa u} K_X(u) du = \sinh(\kappa(1 - X)) / \sinh \kappa$ . Both sides are analytic on  $|\operatorname{Re} a| < \pi$ .  $\square$

**Lemma 4.2.** For  $0 < X < 1$ ,  $h_\lambda(X + iY) = \int_{\mathbb{R}} (w - \lambda)_+ K_X(Y - w) dw$ ;  $h_\lambda$  extends continuously to  $S$  with  $h_\lambda(iY) = (Y - \lambda)_+$  and  $h_\lambda(1 + iY) = 0$ .

*Proof.* The right side  $\Psi$  is harmonic ( $K_X(Y - w) = \frac{1}{2} \operatorname{Re} \cot(\pi(z - iw)/2)$  with  $z = X + iY$ ). For  $Y < \lambda$  expand the kernel by Lemma 4.1 and integrate termwise:

$$\Psi = \sum_n \frac{\sin(n\pi X) e^{n\pi(Y-\lambda)}}{(n\pi)^2} = -\frac{1}{\pi^2} \operatorname{Im} \sum_n \frac{(e^{\pi(Y-\lambda)} e^{-i\pi X})^n}{n^2} = h_\lambda.$$

Two harmonic functions on the connected strip that agree on an open set agree everywhere. As  $X \rightarrow 0$ ,  $K_X(Y - \cdot)$  is an approximate identity at  $Y$ ; as  $X \rightarrow 1$ ,  $K_X \rightarrow 0$  with exponential tails.  $\square$

For  $\nu = x + iy \in S$  let  $\omega_\nu$  be the measure  $K_x(y - w) dw$  ( $0 < x < 1$ ),  $\delta_y$  ( $x = 0$ ) or 0 ( $x = 1$ ). Then  $h_\lambda(\nu) = \int (w - \lambda)_+ \omega_\nu(dw)$  and  $\int e^{aw} \omega_\nu(dw) = e^{ay} \sin(a(1-x))/\sin a$  ( $|\operatorname{Re} a| < \pi$ ).

**Theorem 4.3** (Theorem 3 of the article, with the defect formula of Theorem 4). *For every  $\lambda \in \mathbb{R}$ ,*

$$\sum_{\nu \in \operatorname{spec}(B+ig)} h_\lambda(\nu) - \operatorname{Tr}[(P(g - \lambda)P)_+] = V(\lambda) := \int_0^1 \int_{\mathbb{R}} K_{1-\tau}(s - \lambda) F(s, \tau) ds d\tau \geq 0,$$

*with eigenvalues counted with algebraic multiplicity. Equality holds for one (equivalently, every)  $\lambda$  if and only if  $[B, g] = 0$ .*

*Proof.* The eigenvalues  $\nu_k = x_k + iy_k$  of  $X = B + ig$  satisfy  $x_k \in [0, 1]$ , since  $\operatorname{Re} \langle \psi, X\psi \rangle = \langle \psi, B\psi \rangle$ . Let  $\rho = \sum_k \omega_{\nu_k}$  and  $\sigma = \sum_{\mu \in \operatorname{spec}(PgP|_{\operatorname{ran} P})} \delta_\mu$ ; the left side is  $\int (w - \lambda)_+ d(\rho - \sigma)(w)$ . (a) For real  $0 < |a| < \pi$ ,  $\hat{\rho}(a) := \int e^{aw} \rho(dw) = \sum_k e^{ay_k} \sin(a(1-x_k))/\sin a = [\operatorname{Tr} e^{ag+iaP} - \operatorname{Tr} e^{ag-iaP}]/(2i \sin a)$ , because  $e^{ag+iaP} = e^{-ia(X-1)}$  and  $e^{ag-iaP} = e^{ia(X^\dagger-1)}$ . Also  $\hat{\sigma}(a) = \operatorname{Tr}_P e^{aPgP}$ . The  $B$ -terms cancel in  $\mathcal{D}(a, -ia) - \mathcal{D}(a, ia) = 2i \sin a [\hat{\rho}(a) - \hat{\sigma}(a)]$ , and by Theorem 3.1 at  $t = \mp ia$  this equals  $2ia^2 \iint e^{as} \sin(a\tau) F ds d\tau$ . With Lemma 4.1 ( $X = 1 - \tau$ ) and Tonelli,

$$\hat{\rho}(a) - \hat{\sigma}(a) = a^2 \iint e^{as} \frac{\sin a\tau}{\sin a} F ds d\tau = a^2 \int_{\mathbb{R}} e^{aw} V(w) dw.$$

$V$  is finite, continuous and  $O(e^{-\pi|w|})$  ( $\int K_{1-\tau} = \tau$  and  $F \leq C/\sqrt{\tau(1-\tau)}$ ). (b) Hence  $\rho$  and  $\sigma$  have equal mass and equal first moment. Put  $W(w) = \int (w - u)_+ d(\rho - \sigma)(u)$ ;  $W$  is continuous,  $O(e^{-\pi|w|})$  at both ends (by the moment identities), and  $W'' = \rho - \sigma$ . Two integrations by parts give  $\int e^{aw} W = a^{-2}(\hat{\rho} - \hat{\sigma}) = \int e^{aw} V$  for real  $0 < |a| < \pi$ ; both sides are analytic on  $|\operatorname{Re} a| < \pi$ , so  $W$  and  $V$  have the same Fourier transform and  $W = V$ . (c) Since  $(u - \lambda)_+ = (\lambda - u)_+ + (u - \lambda)$  and  $\rho - \sigma$  has zero mass and first moment, the left side equals  $W(\lambda) = V(\lambda) \geq 0$ . (d)  $K_{1-\tau} > 0$  for  $0 < \tau < 1$  and  $F \geq 0$  is continuous, so  $V(\lambda) = 0$  forces  $F \equiv 0$  on  $\mathbb{R} \times (0, 1)$ , then  $\mathcal{D} \equiv 0$  by Theorem 3.1, then  $\|BgP\|_F^2 = 0$  by Remark 3.6, i.e.  $[B, g] = 0$ . Conversely, if  $[B, g] = 0$  then  $p = q$ ,  $F = 0$  and  $V = 0$ .  $\square$

**Remark 4.4** (convex order). By (b)–(c), Theorem 4.3 for all  $\lambda$  says that  $\sigma$  (the spectrum of the compression  $PgP$ ) is dominated by  $\rho$  (the eigenvalues of  $B + ig$  swept to the left edge by harmonic measure) in convex order:  $\sum_\mu f(\mu) \leq \int f d\rho$  for every convex  $f$ . Applied to  $(P, g)$  it gives the mirror statement for  $BgB$  and the right edge.

**Remark 4.5** (a second proof in special cases). For  $\operatorname{rank} B \leq 1$ , for  $\operatorname{corank} B \leq 3$  and for all  $M \leq 5$ , Theorem 4.3 also follows from a one-dimensional reduction (the compression eigenvalues of a rank-one perturbation are quantiles of a Cauchy-type distribution) combined with an argument-principle rigidity of the critical points of the defect. Earlier, the cases  $M \leq 4$  and  $\operatorname{corank} B \leq 2$  had been proved by a geometric argument based on Pedoe's inequality for polygons and, independently for  $M \leq 4$ , by a Bellman-function argument. These proofs are documented in the repository[25].

## 5. THE CONTINUUM THEOREM FOR STEP FIELDS

Let  $\mathbb{T} = \mathbb{R}/2\pi\mathbb{Z}$ ,  $dm = d\theta/2\pi$ , and let  $\Phi_c(\alpha, \beta) = \frac{1}{2\pi^2}[\text{Cl}_2(2\pi\alpha) + \text{Cl}_2(2\pi\beta) + \text{Cl}_2(2\pi(1-\alpha-\beta))]$ , where  $\text{Cl}_2(\theta) = \sum_{n \geq 1} \sin(n\theta)/n^2$  is the Clausen function. For  $\alpha = \beta = \frac{1}{4}$ ,  $\Phi_c = G/\pi^2$  with  $G$  Catalan's constant. We need the theorem only for step fields, for which every step can be made explicit.

**Theorem 5.1** (continuum theorem for step fields). *Let  $t_0, \dots, t_{n-1}$  be distinct points of  $\mathbb{T}$  in cyclic order,  $J_j = (t_j, t_{j+1})$ , and let  $B_j$  be orthogonal projections and  $0 \leq A_j \leq \mathbf{1} - B_j$  on  $\mathbb{C}^M$ . Let  $B(\theta) = B_j$ ,  $A(\theta) = A_j$  on  $J_j$ , and assume  $\int B dm = \beta \mathbf{1}$ . Let  $g$  be the conjugate function of  $B$ ,*

$$g(\theta) = \frac{1}{\pi} \sum_j (B_j - B_{j-1}) \log \left| \sin \frac{\theta - t_j}{2} \right| \quad (\theta \notin \{t_j\}).$$

*Then, with  $\alpha = \int \tau(A) dm$ ,  $\int \tau(Ag) dm \leq \Phi_c(\alpha, \beta)$ . More precisely, for every  $\lambda$ ,*

$$(5.1) \quad \int \tau(Ag) dm \leq \lambda\alpha + \int \tau((P(g - \lambda)P)_+) dm \leq \lambda\alpha + h_\lambda(\beta), \quad P = \mathbf{1} - B,$$

*and if  $\alpha = \beta = \frac{1}{4}$  and equality holds throughout at  $\lambda^* = \log 2/(2\pi)$ , then  $[B(\theta), g(\theta)] = 0$  for almost every  $\theta$ .*

*Proof. Herglotz function.*  $H(z) = \int \frac{e^{i\phi} + z}{e^{i\phi} - z} B(\phi) dm(\phi) = \beta \mathbf{1} + \frac{i}{\pi} \sum_j (B_j - B_{j-1}) \text{Log}(1 - ze^{-it_j})$  is analytic on the disc,  $H(0) = \beta \mathbf{1}$ , and extends continuously to the closed disc minus  $\{e^{it_j}\}$  with boundary values  $B(\theta) + ig(\theta)$ . For  $r \geq \frac{1}{2}$ ,  $\|H(re^{i\theta})\| \leq C_0 + \frac{2}{\pi} \sum_j |\log |\sin \frac{\theta - t_j}{2}||$ , a function in every  $L^p$ . *Spectrum in the strip.*  $\langle \psi, \text{Re } H(z)\psi \rangle = \int P_z \langle \psi, B\psi \rangle dm \in (0, 1)$  for unit  $\psi$  unless  $B(\phi)\psi = 0$  a.e. or  $B(\phi)\psi = \psi$  a.e.; after removing the common kernel and range of the  $B_j$  (on which the statement is trivial) the numerical range of  $H(z)$ , hence its spectrum, lies in the open strip. *Harmonicity.*  $\phi_\lambda(w) = \frac{i}{\pi^2} \text{Li}_2(e^{-i\pi w - \pi\lambda})$  is analytic on the open strip with  $\text{Re } \phi_\lambda = h_\lambda$ , so  $u_\lambda(z) = \text{Re Tr } \phi_\lambda(H(z)) = \sum_{\nu \in \text{spec } H(z)} h_\lambda(\nu)$  is harmonic, with  $u_\lambda(0) = Mh_\lambda(\beta)$ . *Boundary passage.*  $0 \leq h_\lambda(x + iy) \leq |y| + |\lambda| + C_1$ , so  $u_\lambda(re^{i\theta})$  is dominated by an integrable function; for  $\theta \notin \{t_j\}$  it converges to  $u_\lambda^*(\theta) = \sum_{\nu \in \text{spec}(B+ig)(\theta)} h_\lambda(\nu)$ . By the mean value property and dominated convergence,  $\int u_\lambda^* dm = Mh_\lambda(\beta)$ . *Chain.* For  $\theta \notin \{t_j\}$ , with  $X = P(g - \lambda)P$ :  $\tau(A(g - \lambda)) = \tau(AX) \leq \tau(AX_+) \leq \tau(X_+)$ , and by Theorem 4.3,  $\tau(X_+) \leq \frac{1}{M} u_\lambda^*(\theta)$ . Integrating gives (5.1). *Legendre identity.*  $\min_\lambda [\lambda\alpha + h_\lambda(\beta)] = \Phi_c(\alpha, \beta)$ , attained at  $\lambda = \frac{1}{\pi} \log \frac{\sin \pi(\alpha + \beta)}{\sin \pi\alpha}$ ; for  $\alpha = \beta = \frac{1}{4}$  this is  $\lambda^*$ . *Equality.* If both inequalities in (5.1) are equalities at  $\lambda^*$ , then the non-negative integrand  $\frac{1}{M} u_{\lambda^*}^* - \tau((P(g - \lambda^*)P)_+)$  vanishes a.e., i.e. Theorem 4.3 holds with equality at  $\lambda^*$  for a.e.  $\theta$ , and its equality clause gives  $[B(\theta), g(\theta)] = 0$  a.e.  $\square$

**Remark 5.2.** With only the trace constraint  $\int \tau(B) = \beta$ , the centre value is  $\sum_i h_\lambda(\beta_i) \leq Mh_\lambda(\beta)$  by concavity of  $x \mapsto h_\lambda(x)$  on  $[0, 1]$  (the  $y$ -convexity of  $h_\lambda$  and harmonicity), and the same bound follows. The operator constraint holds in all applications below.

6. CELL EMBEDDINGS AND THE REDUCTION OF THEOREM 1 TO CONE<sub>d</sub>

Let  $\Delta_d = \{\ell \in \mathbb{R}^d : \ell_r \geq 0, \sum_r \ell_r = d\}$ ; extend  $\ell_x = \ell_{x \bmod d}$  to  $x \in \mathbb{Z}_N$ , put  $L_0 = 0$ ,  $L_{x+1} = L_x + \ell_x$  (so  $L_N = N$ ) and  $I_x = [L_x, L_{x+1}) \subset \mathbb{R}/N\mathbb{Z}$ . For a 27B configuration let  $\tilde{A}, \tilde{B}$  be the step fields equal to  $A_x, B_x$  on  $I_x$ , and

$$Q^\ell = \iint \tau(\tilde{A}(\xi)\tilde{B}(\eta)) \cot \frac{\pi(\xi - \eta)}{N} d\xi d\eta = \sum_{x,y} K^\ell(x, y) \tau(A_x B_y),$$

with  $K^\ell(x, y) = \int_{I_x} \int_{I_y} \cot \frac{\pi(\xi-\eta)}{N} d\xi d\eta$ . With  $G(u) = -\frac{N^2}{2\pi^2} \text{Cl}_2(2\pi u/N)$  (so  $G''(u) = \cot(\pi u/N)$ ),  $K^\ell(x, y)$  is the mixed second difference of  $G$  over the cell endpoints.

**Proposition 6.1** (cell inequalities). *For every  $\ell \in \Delta_d$  and every 27B configuration,  $Q^\ell \leq N^2 \Phi_c(\frac{1}{4}, \frac{1}{4})$ , with equality for the DKZ configuration. Averaging over the  $d$  rotations  $Q_x \mapsto Q_{x-c}$ ,*

$$(6.1) \quad \frac{1}{d} \sum_{c=0}^{d-1} [Q^\ell(\text{rot}_c Q) - N^2 \Phi_c(\frac{1}{4}, \frac{1}{4})] = \langle u^\ell, \delta \rangle \leq 0, \quad u_m^\ell = \Delta_m^2 \left[ \frac{1}{d} \sum_{r=0}^{d-1} \hat{G}(S_r(m)) \right],$$

where  $S_r(m) = \ell_r + \dots + \ell_{r+m-1}$ ,  $\hat{G}(u) = G(u) + G(2d-u)$ ,  $\Delta_m^2$  is the second difference in  $m$  and  $1 \leq m \leq d-1$ . Moreover  $v_m = \hat{G}''(m)$ .

*Proof.* Rescale  $\mathbb{R}/N\mathbb{Z}$  to  $\mathbb{T}$ . Since  $\ell$  depends only on the residue modulo  $d$  and  $\{Q_{r+jd}\}_j$  is a projective measurement for each  $r$ ,  $\int \tilde{B} dm = \frac{1}{N} \sum_r \ell_r \sum_j Q_{r+d+jd} = \frac{d}{N} \mathbf{1} = \frac{1}{4} \mathbf{1}$ , and likewise  $\int \tilde{A} dm = \frac{1}{4} \mathbf{1}$ ; also  $\tilde{A}\tilde{B} = 0$ . Zero-length cells carry no field. Theorem 5.1 (with  $\alpha = \beta = \frac{1}{4}$ ;  $g$  is the conjugate function of the step field) gives  $Q^\ell/N^2 = \int \tau(\tilde{A}g) dm \leq \Phi_c(\frac{1}{4}, \frac{1}{4})$ . For DKZ, the windows  $[c, c+d)$  and  $[c-d, c)$  occupy adjacent arcs of length  $d$ , whose continuum value is  $N^2 \Phi_c$  (Stein–Weiss extremal). Rotations preserve the class of 27B configurations. The kernel  $K^\ell$  is invariant under  $(x, y) \mapsto (x+d, y+d)$ , so the rotation average is  $\sum_m \tilde{K}(m) T(m)$  with  $\tilde{K}(m) = \frac{1}{d} \sum_{r < d} K^\ell(r+m, r)$ , odd in  $m$ ; folding with  $N(2d-m) = N(m)$ ,  $N(d) = d$  as in Lemma 2.2 gives (6.1) with  $u_m^\ell = \tilde{K}(m) + \tilde{K}(2d-m)$ . Writing  $K^\ell(r+m, r)$  through window sums,  $\tilde{K}(m) = \Delta^2 P(m)$  with  $P(k) = \frac{1}{d} \sum_r G(S_r(k))$ , and  $S_r(2d-j) = 2d - S_{r+d-j}(j)$  gives the window-sum form. Finally  $\hat{G}''(u) = \cot \frac{\pi u}{4d} + \tan \frac{\pi u}{4d} = 2 \csc \frac{\pi u}{2d}$ .  $\square$

**Definition 6.2** ( $\text{CONE}_d$ ).  $\text{CONE}_d$  holds if there are a finite positive Borel measure  $\mu$  on  $\Delta_d$  and  $w_s \geq 0$  with  $v = \int_{\Delta_d} u^\ell \mu(d\ell) + \sum_{s=1}^{\lfloor d/2 \rfloor} w_s (e_s + e_{d-s})$ .

*Proof of Theorem 1, given  $\text{CONE}_d$ .* By Lemma 2.2 and Proposition 6.1,  $2(F - F_{\text{DKZ}}) = \langle v, \delta \rangle = \int \langle u^\ell, \delta \rangle \mu(d\ell) + \sum_s w_s (\delta_s + \delta_{d-s}) \leq 0$ . By Theorem 2.1,  $I_d \leq I_{\text{ME}}(d)$  for every projective strategy on every  $\Phi_D$ .  $\square$

**Lemma 6.3** (Dirichlet cells reproduce the singular part). *If  $\ell = d \cdot \text{Dir}(1, \dots, 1)$  (the gaps between  $d$  uniform random points on a circle of length  $d$ ), then for every  $\phi \in C^2(0, d]$  with  $\phi''(s)s^m$  integrable and  $1 \leq m \leq d-1$ ,  $\Delta_m^2 \mathbb{E} \phi(S(m)) = \int_0^d \phi''(s) b_{m,d}(s/d) ds = \frac{d}{d+1} \mathbb{E} \phi''(dX)$  with  $X \sim \text{Beta}(m+1, d-m+1)$ ,  $b_{m,d}(x) = \binom{d}{m} x^m (1-x)^{d-m}$ . In particular for  $\phi''(s) = 1/s$  the result is exactly  $1/m$ .*

*Proof.*  $S(m)/d \sim \text{Beta}(m, d-m)$  has density  $\frac{d-1}{d} b_{m-1, d-2}(s/d)$ , and  $b''_{m,d} = d(d-1)(b_{m-2, d-2} - 2b_{m-1, d-2} + b_{m, d-2})$ ; two integrations by parts, in which the point masses of  $S(0) = 0$  and  $S(d) = d$  cancel the boundary terms, give the formula.  $\square$

Lemma 6.3 explains the construction of the certificates: the expectation of  $u^\ell$  under uniform Dirichlet cells is a Bernstein–Durrmeyer smoothing of  $\hat{G}''$  which reproduces the pole  $2 \csc(\pi u/2d) \sim 4d/(\pi u)$  exactly, and the residual comes only from the image poles at distance  $\geq d$ . It is smooth and of relative order  $1/d$ , and it must be corrected by non-exchangeable cell distributions.

## 7. $\text{CONE}_d$ FOR EVERY $d$ (COMPUTER-ASSISTED)

**Theorem 7.1** (computer-assisted).  *$\text{CONE}_d$  holds for every  $d \geq 2$ . For  $d \leq 200$ ,  $\mu$  is a finite positive combination of point masses at rational cell vectors with all cells  $\geq \frac{1}{16}$ , and  $w = 0$ . For  $d \geq 201$ ,  $\mu$  is the law of a Gaussian-modulated Dirichlet process, which charges only the open simplex.*

**7.1.  $2 \leq d \leq 200$ : exact rational cells.** For each  $d$ ,  $K$  cell vectors  $\ell^{(k)} = n^{(k)}/16$  with integers  $n_r^{(k)} \geq 1$ ,  $\sum_r n_r^{(k)} = 16d$  ( $3.5d \leq K \leq 6.3d$ ) are given. All window sums are multiples of  $\frac{1}{16}$ , so  $\hat{G}$  is needed only at  $j/16$ :  $\hat{G}(j/16) = -\frac{N^2}{2\pi^2}[\text{Cl}_2(\pi j/K_g) + \text{Cl}_2(\pi(K_g - j)/K_g)]$ ,  $K_g = 32d$ . Each value is enclosed in 160-bit interval arithmetic by  $\text{Cl}_2(t) = t - t \log t + \sum_{n \leq 70} \frac{|B_{2n}| t^{2n+1}}{2n(2n+1)!} + [0, \epsilon_{70}(t)]$  ( $0 < t \leq \pi$ ) with exact Bernoulli numbers and an explicit tail bound. A floating-point linear program gives weights  $\tilde{\lambda} > 0$ ; the exact solution  $\lambda = \tilde{\lambda} + U^T y$  of  $U\lambda = v$  is enclosed through  $|y| \leq |R \text{res}|/(1 - \|I - RUU^T\|)$  with an approximate inverse  $R$ . Every weight is certified positive ( $d \cdot \min_k \lambda_k \in [0.0816, 0.2572]$ ; smallest  $4.25 \times 10^{-4}$ , at  $d = 198$ ), and a complete re-verification of all 199 certificates in interval arithmetic passed. The enclosures of the Clausen function were compared with an independent 63-point high-precision evaluation. An independent scan of all 199 certificates confirmed the cell structure and the positivity of the weights, and independent floating-point re-evaluations of the identity at selected  $d$  (with 50-digit Clausen values and, for  $d \leq 10$ , the original cell-pair kernel instead of the window-sum form) agree to the precision of the stored weights. For  $2 \leq d \leq 20$  the same certificates are also re-evaluated by the Lean kernel in exact rational interval arithmetic (Section 10).

**7.2.  $201 \leq d \leq 2000$ : Gaussian-modulated Dirichlet cells. Potentials.** For a cell process put  $P(m) = \mathbb{E}_d^1 \sum_r \hat{G}(S_r(m))$ , so that  $u = \Delta^2 P$ , and let  $P_v$  solve  $\Delta^2 P_v = v$  on  $1, \dots, d-1$  with the same boundary values. Write  $X_A(m) = \frac{1}{2}(X(m) - X(d-m))$  and  $X_S(m) = \frac{1}{2}(X(m) + X(d-m))$ .

*Lemma R1 (reduction).* If (i)  $P_A = (P_v)_A$  and (ii)  $W = (P_v - P)_S$  has  $\Delta^2 W \geq 0$  on  $1, \dots, d-1$ , then  $\text{CONE}_d$  holds (with  $w = \Delta^2 W$ , a non-negative combination of the  $e_s + e_{d-s}$ ).

*Lemma R2 (exact singular part).*  $\hat{G} = \hat{G}_s + \hat{G}_r$  with  $\hat{G}_s(u) = \frac{4d}{\pi} u \log \frac{u}{d}$  and  $\hat{G}_r(u) = d^2 g_r(u/d)$ ,  $g_r$  analytic on  $|x| < 2$ . For Dirichlet cells the singular part of  $P$  equals that of  $P_v$  exactly,  $\frac{4d}{\pi}[m\psi(m) + 1 - \frac{m}{d} - m\psi(d)]$  ( $\psi$  the digamma function), so only the analytic  $g_r$  enters condition (i).

*Process.* Let  $\Phi : \{0, \dots, d\} \rightarrow [0, \infty)$  be symmetric with  $\Phi(0) = \Phi(d) = 0$  and Fourier coefficients  $a_k(\Phi) = -\frac{2w_k}{d} \sum_m \Phi(m) \cos \frac{2\pi km}{d} > 0$  ( $1 \leq k \leq d/2$ ), where  $w_k = 1$  for  $k < d/2$  and  $w_{d/2} = \frac{1}{2}$ . Then  $c(j) = \frac{1}{2}\Delta^2 \Phi(j)$  is a positive-definite function on  $\mathbb{Z}_d$  with zero sum, and the centred stationary Gaussian vector  $\eta$  with covariance  $c$  satisfies  $\sum_r \eta_r = 0$  and  $\text{Var}(\eta_0 + \dots + \eta_{m-1}) = \Phi(m)$ . On  $E = \{\max_r |\eta_r| < \frac{3}{4}\}$  take  $\ell = d \cdot \text{Dir}(1 + \eta)$ , on  $E^c$  the uniform Dirichlet law.

*Lemmas R3, R4 (decoupling).* By the aggregation property of the Dirichlet law,  $P(m) = \mathbb{E}[\mathcal{F}(m+Z(m)); E] + \mathbb{P}(E^c)\mathcal{F}(m)$  with  $Z(m) \sim N(0, \Phi(m))$  and  $\mathcal{F}(b) = \mathbb{E}\hat{G}(dX_b)$ ,  $X_b \sim \text{Beta}(b, d-b)$ . Up to an error bounded by  $2d \exp(-9/(32\Phi(1)))$  times explicit derivative bounds, condition (i) decouples into one scalar equation  $\mathcal{H}_m(\Phi(m)) = \tau_m$  per  $1 \leq m < d/2$ .

*Lemma B (fixed point).* If on a box  $Q = \prod_m [\text{lo}_m, \text{hi}_m]$  all  $a_k > 0$ , the coupling errors are at most  $2\epsilon_t$ , and  $\mathcal{H}_m(\text{lo}_m) < \tau_m - 2\epsilon_t$ ,  $\mathcal{H}_m(\text{hi}_m) > \tau_m + 2\epsilon_t$ , then the Poincaré–Miranda theorem[18] gives  $\Phi^\# \in Q$  for which (i) holds exactly. The two sign conditions follow from a certified root enclosure  $[a_m, b_m]$  and a certified lower bound  $c$  on  $\mathcal{H}'_m$  on  $[a_m - \rho, b_m + \rho]$  with  $c\rho \geq 2\epsilon_t$ .

*Certificates.* For each  $201 \leq d \leq 2000$  one run of a single checker certifies all hypotheses of Lemma B and condition (ii) of Lemma R1. The run uses one precision (160-bit interval arithmetic throughout) and one code version, whose SHA-256 hash is stored in every certificate. It encloses  $\mathcal{F}^{(2j)}$  ( $j \leq 7$ ) at the integers (polygamma values through zeta values and harmonic sums, and the regular part through exact truncated Pochhammer products with a rigorous tail), the targets  $\tau_m$ , and the decoupled solution  $\Phi_*$  (exact order-14 Gaussian expansion and order-16 Lagrange remainder); it finds root boxes  $[a_m, b_m]$  with a certified sign change and fixes one Poincaré–Miranda box  $Q \supseteq \prod_m [a_m - r, b_m + r]$ , with a radius  $r = 2^e$  chosen from the tail bound. All remaining conditions are certified on this widened box: all  $a_k > 0$ ; a lower bound  $c$  for  $\mathcal{H}'_m$  with  $c\rho > 2\epsilon_t$ ; the two sign conditions, also directly at the faces of  $Q$ ; a lower bound larger than  $4\epsilon_t$ .

for the second differences of the symmetric part of  $P_v$  minus the Gaussian average, which gives (ii) for the exact process; and the tail bound, in logarithms. Factorials, binomial coefficients and Bernoulli numbers enter as exact integers and rationals, the zeta values are enclosed by the Euler–Maclaurin formula with a rigorous remainder, and no decision is taken on floating-point numbers. An audit script re-checks every condition from the exact decimal values stored in the certificates: 1800 of 1800 pass. The smallest certified values over all  $d$  are  $d^2 \min_k a_k \geq 0.06319$  (even  $d$ ; at  $d = 2000$ ) and  $\geq 0.12638$  (odd  $d$ ),  $d \min_m \Delta^2 W \geq 0.660$ , a ratio of the certified derivative bound to the required one of 4.0009 (at least 1 is needed), face margins at least  $10^{16.7} \cdot 2\epsilon_t$ ,  $\epsilon_t \leq 10^{-31.16}$  and  $\mathbb{P}(E^c) \leq 4.2 \times 10^{-37}$ . (Earlier runs at 100 bits for  $d \leq 600$  gave lower apparent margins, 0.0285 and 0.0798; they are superseded.)

**7.3.  $d \geq 2001$ : analytic proof with interval Taylor models.** Let  $\epsilon = 1/d$ . A smooth model  $\Phi_e(b) = \epsilon b^2 u_e(b)$  is defined,  $u_e$  being the unique root in  $(0, 1)$  of a quartic  $\sum_{j=1}^4 \text{Ch}_j(b) u^j = \hat{\tau}_e(b)$  with explicit coefficients (Binet-type functions of the digamma derivatives and the Taylor coefficients of  $g_r$ , and an Euler–Maclaurin interpolant of  $P_v$ ). Positivity of the  $a_k$  reduces, by a Fejér–Pólya argument for the convex sequence  $a(m) = \frac{1}{6} - f(m)$ , to the pointwise bounds  $\Delta^4(\Phi_e - F_A)(1) \geq -\gamma\epsilon^2$  and  $\Delta^4(\Phi_e - F_A)(m) \geq \beta\epsilon^3$  ( $2 \leq m \leq d/2$ ), where  $F_A$  is an explicit boundary layer, together with  $|\Phi_* - \Phi_e| \leq \delta_{\max}$ . For instance,  $\gamma = 3$ ,  $\beta = \frac{1}{4}$  and  $\delta_{\max} = \epsilon^2/20$  would already give  $a_k \geq 0.011 w_k \epsilon^2 > 0$  for every  $d \geq 2001$ ; this is the sense of the sufficient level  $\frac{1}{4}$  in Fig. 5d of the article. The bounds actually certified, by interval Taylor models in  $(x, \epsilon)$ ,  $x = b/d$ , on boxes covering every  $d \geq 2001$  simultaneously, are much better:

| quantity                                    | region                                                       | certified bound                                                                                       |
|---------------------------------------------|--------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| normalized fourth difference,<br>outer zone | $0.011 \leq x \leq 0.501$ ,<br>$0 \leq \epsilon \leq 1/2001$ | $\geq 7.419$                                                                                          |
| normalized fourth difference,<br>inner zone | $x \leq 0.013$ , all positions $b \geq 1$                    | $\geq 6.022$                                                                                          |
| first two stencil positions                 | $m = 1; m = 2$                                               | $\geq -2.765192 \epsilon^2$ ;<br>$\geq 206.884 \epsilon^3$                                            |
| model error                                 | all $m$                                                      | $ \Phi_* - \Phi_e  \leq 6.6468 \times 10^{-4} \epsilon^2$                                             |
| Fourier coefficients                        | all $k$                                                      | $a_k \geq 0.1186944 w_k \epsilon^2$                                                                   |
| symmetric slack                             | all $m$                                                      | $\Delta^2 W \geq 0.620154 \epsilon$                                                                   |
| higher Gaussian orders                      | all $m$                                                      | $ \mathbf{G} - \mathbf{G}_4  \leq 138.103 \epsilon^3$ ,<br>$ \partial_t \mathbf{G}_4  \leq 11.5703 d$ |
| Fejér–Pólya constant                        |                                                              | $\min \mathcal{A} \geq 0.1054878997577$                                                               |
| tail and fixed-point conditions             | all $d \geq 2001$                                            | logarithmic margins $> 800$                                                                           |

Here  $\mathbf{G}(b, t)$  is the truncated Gaussian average of  $\mathcal{F}(b + \cdot) - \mathcal{F}(b)$  with variance  $t$ ,  $\mathbf{G}_4$  its expansion to order  $t^4$ , and  $\mathcal{A}(\theta) = a(0) + 2 \sum_{m \geq 1} a(m) \cos m\theta$ . All constants, including these, are computed in interval arithmetic. A rigour pass completed the written proofs of the model identities (one of them, the first-order identity of the inner zone, is now proved in full and was checked symbolically and numerically to  $6 \times 10^{-25}$ ) and corrected several code-level enclosure defects, one of which had left a sliver of size about  $10^{-20}$  of the parameter range uncovered at  $d = 2001$ ; no conclusion changed. An audit script checks the coverage of all boxes and recomputes every margin from the logs. Non-rigorous cross-checks (for example  $d^2 \Delta^4(1) = -2.7514$  and  $d^3 \Delta^4(m) \approx 10$  at  $d = 4000$ ) agree with the certified bounds.

*Remark 7.2 (coverage).* All 1800 single-run certificates for  $201 \leq d \leq 2000$ , one per  $d$  and all from the same code version, pass the audit, the smallest derivative ratio being 4.0009; the audit for  $d \geq 2001$  reports every box and every margin as certified. A single command re-checks all three regimes (the 199 certificates for  $d \leq 200$  against their verification records, the 1800 certificates for

201  $\leq d \leq 2000$ , and every box and margin for  $d \geq 2001$ ) and reports  $\text{CONE}_d$  certified for every  $d \geq 2$ . Fig. 5 of the article is generated directly from these certificate files.

*Remark 7.3* (finite form). For  $d \geq 201$  the certificates give  $\text{CONE}_d$  in integral form, with a measure  $\mu$  carried by the open simplex. A finite form,  $v = \sum_j \alpha_j u^{\ell_j}$  with  $\alpha_j \geq 0$  and at least one  $\ell_j$  with all cells positive and  $\alpha_j > 0$ , follows from it: the barycentre of  $\mu$  lies in the convex hull of  $\{u^\ell : \ell \text{ in the open simplex}\}$ , Carathéodory's theorem reduces it to at most  $d$  points, and each trivial direction  $e_s + e_{d-s}$  is a positive multiple of the cell vector of a cell sequence supported on two residues. This finite form is the input of the Lean formalisation (Section 10).

## 8. PROOF OF THEOREM 2 (RIGIDITY)

Let a projective strategy  $R$  on  $\Phi_D$  attain  $I_{\text{ME}}(d)$ ; let  $V$  be its reduced family and  $Q$  its 27B configuration on  $\mathbb{C}^M$ . Then  $\langle v, \delta \rangle = 0$  (Theorem 2.1, Lemma 2.2).

*Step 1.* In the representation of Theorem 7.1 every term is  $\leq 0$  (Proposition 6.1, Lemma 2.2), so  $\langle u^\ell, \delta \rangle = 0$  for  $\mu$ -almost every  $\ell$ . Since  $\mu$  charges  $\Delta_d^+ = \{\ell : \ell_r > 0 \ \forall r\}$  (all cells  $\geq \frac{1}{16}$  for  $d \leq 200$ ; Dirichlet laws with parameters  $\geq \frac{1}{4}$  for  $d > 200$ ), there is  $\ell^* \in \Delta_d^+$  with  $\langle u^{\ell^*}, \delta \rangle = 0$ . All terms of the rotation average (6.1) are  $\leq 0$  with mean 0, so  $Q^{\ell^*}(Q) = N^2 \Phi_c(\frac{1}{4}, \frac{1}{4})$ . (The uniform cell vector would not suffice:  $u^{(1, \dots, 1)}$  is not proportional to  $v$ .)

*Step 2.* By the equality clause of Theorem 5.1,  $[B(\theta), g(\theta)] = 0$  for almost every  $\theta$ , where  $B$  is the step field of  $Q$  for the cells of  $\ell^*$ .

**Lemma 8.1** (residue lemma). *Let  $t_0, \dots, t_{n-1} \in \mathbb{R}$  be distinct modulo  $2\pi$  and  $J_j \in M_M(\mathbb{C})$ . If  $\sum_j J_j \cot \frac{\theta - t_j}{2} = 0$  on a non-empty open interval avoiding  $\bigcup_j (t_j + 2\pi\mathbb{Z})$ , then all  $J_j = 0$ .*

*Proof.* The function  $\sum_j J_j \cot \frac{z - t_j}{2}$  is holomorphic on the connected set  $\mathbb{C} \setminus \bigcup_j (t_j + 2\pi\mathbb{Z})$  and vanishes on an interval, hence everywhere; its residue at  $t_j$  is  $2J_j$ .  $\square$

*Step 3.* On each open arc  $J_x$  the field  $g$  is real-analytic and  $[B_x, g(\theta)] = 0$  a.e., hence identically. Differentiating,  $\sum_j [B_x, B_j - B_{j-1}] \cot \frac{\theta - t_j}{2} = 0$  on  $J_x$ . All cells of  $\ell^*$  have positive length, so the  $t_j$  are distinct, and Lemma 8.1 gives  $[B_x, B_j] = [B_x, B_{j-1}]$  for all  $j$ ; so  $[B_x, B_j] = [B_x, B_x] = 0$ . Hence all  $Q_x$  commute. (Cells of zero length would hide the corresponding measurement; this is why Step 1 needs  $\ell^* \in \Delta_d^+$ .)

*Step 4.* Since  $E_k = \sum_a i^a Q_{k-da}$ , the  $V_k$  commute. By Theorem B of ref. [23] every joint eigenvector of the  $V_k$  carries a one-step configuration, and by Corollary C of ref. [23] the strategy  $R$ , a direct summand of  $R^V$ , is unitarily equivalent to  $R^{(0)} \otimes \mathbf{1}_K$  with  $d \cdot \dim K = D$ . Translating back to measurements, a unitary  $u : \mathbb{C}^D \rightarrow \mathbb{C}^d \otimes \mathbb{C}^K$  with  $u R_i u^\dagger = R_i^{\text{DKZ}} \otimes \mathbf{1}_K$  gives  $u P_a^x u^\dagger = P_a^{x, \text{DKZ}} \otimes \mathbf{1}$  and  $\bar{u} Q_b^y \bar{u}^\dagger = Q_b^{y, \text{DKZ}} \otimes \mathbf{1}$ , i.e. the local unitary  $u \otimes \bar{u}$  maps the strategy to  $\text{DKZ} \otimes \mathbf{1}_K$  and  $\Phi_D$  to  $\Phi_d \otimes \Phi_K$ . The  $R_i^{\text{DKZ}}$  generate  $M_d(\mathbb{C})$ , so  $u$  is unique up to  $\mathbf{1}_d \otimes U(K)$ . If  $d \nmid D$ , compactness of the set of strategies on  $\Phi_D$  shows that the maximum is attained and hence is strictly below  $I_{\text{ME}}(d)$ .  $\square$

## 9. THE NOISE AND KULLBACK–LEIBLER CLAUSES

Part B of Problem 27 also asks to show that the DKZ measurements realise the highest resistance of the violation to noise, and the best discrimination against local realism in the sense of the Kullback–Leibler divergence[28]. This section gives the statements and proofs (sketches for the Kullback–Leibler clause). The settings are  $x, y \in \{1, 2\}$  and the outcomes  $a, b \in \mathbb{Z}_d$ , as in the CGLMP expression of Section 2, and  $\text{DKZ}_d$  denotes the DKZ strategy on  $\Phi_d$ , with the bases  $|k\rangle_{A,x} = d^{-1/2} \sum_j e^{-2\pi i j(k+\alpha_x)/d} |j\rangle$  and  $|l\rangle_{B,y} = d^{-1/2} \sum_j e^{2\pi i j(l-\beta_y)/d} |j\rangle$ ,  $\alpha = (0, \frac{1}{2})$ ,  $\beta = (\frac{1}{4}, -\frac{1}{4})$ .

Certificates, verification programs and the reports of the independent verifications are in the directory `complete-resolution/` of the repository[25]; its file `LEDGER.md` records the wording of the problem[16] and the verdict on every clause.

**9.1. Critical visibilities.** A behaviour is a family  $p = (p(\cdot, \cdot | x, y))_{x,y}$  of probability distributions on  $\mathbb{Z}_d \times \mathbb{Z}_d$ . The local polytope  $\mathcal{L}_d$  is the convex hull of the deterministic behaviours  $\delta_\lambda(a, b | x, y) = [a = a_x][b = b_y]$ ,  $\lambda = (a_1, a_2, b_1, b_2) \in \mathbb{Z}_d^4$ . Let  $p_0(a, b | x, y) = d^{-2}$  (uniformly random outcomes),  $p_v = vp + (1 - v)p_0$  and  $v_c(p) = \max\{v \in [0, 1] : p_v \in \mathcal{L}_d\}$ ;  $p_v$  is local exactly for  $v \leq v_c(p)$ , and a smaller  $v_c$  means a higher resistance to noise. Relabellings of outcomes, settings and parties do not change  $v_c$ . Gill measures noise resistance by mixing with “completely random, uniform outcomes”[11] as long as local realism is violated, that is, by  $v_c$ . The noise clause has three readings: (a) noise  $p_0$ , nonlocality detected by the CGLMP inequality; (b) Gill’s literal reading, noise  $p_0$  and nonlocality against all Bell inequalities, that is,  $v_c$ ; (c) white noise on the state,  $v|\Phi_D\rangle\langle\Phi_D| + (1 - v)\mathbf{1}/D^2$ , against all Bell inequalities. In (c) the noise contributes  $\text{Tr } P_a^x \text{Tr } Q_b^y / D^2$ , which equals  $p_0$  when all outcomes have trace  $D/d$ , for example for rank-one measurements on  $\Phi_d$ .

**Proposition 9.1** (CGLMP reading). *Let  $p$  be the behaviour of a projective strategy on  $\Phi_D$ . Then  $I_d(p_v) = vI_d(p)$ , so  $p_v$  violates the CGLMP inequality if and only if  $v > 2/I_d(p)$ , and  $2/I_d(p) \geq 2/I_{\text{ME}}(d)$ , with equality if and only if the strategy is  $\text{DKZ}_d \otimes \mathbf{1}$  up to a local unitary  $u \otimes \bar{u}$ .*

*Proof.*  $I_d$  is linear, and  $I_d(p_0) = 0$  because every probability in the CGLMP expression equals  $1/d$  under  $p_0$ . Apply Theorems 1 and 2.  $\square$

This is formalised in the file `lean/0QP27/CgmpNoise.lean` (Section 10), with the assumptions of Theorems 1 and 2: none for  $2 \leq d \leq 20$ , and  $\text{CONE}_d$  for  $d \geq 21$ .

**9.2. Gill’s literal reading.** Let  $A_1 = \sigma_z$ ,  $A_2 = \sigma_x$ ,  $B_1 = (\sigma_z + \sigma_x)/\sqrt{2}$ ,  $B_2 = (\sigma_z - \sigma_x)/\sqrt{2}$  be the optimal CHSH observables,  $\Pi_\pm^x = \frac{1}{2}(\mathbf{1} \pm A_x)$  and  $\Pi_\pm^y = \frac{1}{2}(\mathbf{1} \pm B_y)$ . For  $d \geq 3$  the strategy  $C_d$  on  $\Phi_d$  is defined as follows. For even  $d$ , write  $\Phi_d = \Phi_2 \otimes \Phi_{d/2}$  (by  $|i\rangle|j\rangle \mapsto |i\frac{d}{2} + j\rangle$ ) and let  $P_0^x = \Pi_+^x \otimes \mathbf{1}$ ,  $P_1^x = \Pi_-^x \otimes \mathbf{1}$ ,  $Q_0^y = \Pi_+^y \otimes \mathbf{1}$ ,  $Q_1^y = \Pi_-^y \otimes \mathbf{1}$ . For odd  $d$ , write  $\mathbb{C}^d = \bigoplus_{k=1}^{(d-1)/2} \text{span}\{|2k-2\rangle, |2k-1\rangle\} \oplus \mathbb{C}|d-1\rangle$  and let  $P_0^x = (\bigoplus_k \Pi_+^x) \oplus \mathbf{1}$ ,  $P_1^x = (\bigoplus_k \Pi_-^x) \oplus 0$ , and  $Q_0^y, Q_1^y$  likewise. In both cases the outcomes  $2, \dots, d-1$  have zero projections. Let

$$v_{\text{even}}(d) = \frac{4(d-1)}{(\sqrt{2}-1)d^2 + 4(d-1)}, \quad v_{\text{odd}}(d) = \frac{4}{(\sqrt{2}-1)d + 4}.$$

**Theorem 9.2** (computer-assisted for  $4 \leq d \leq 9$ ). *(1) For every  $d \geq 3$ ,  $v_c(C_d) = v_{\text{even}}(d)$  for even  $d$  and  $v_c(C_d) = v_{\text{odd}}(d)$  for odd  $d$ . (2) For every  $d \geq 2$ ,  $\frac{1}{2} \leq v_c(\text{DKZ}_d) \leq 2/I_{\text{ME}}(d)$ , and  $v_c(\text{DKZ}_d) \geq 0.69, 0.687, 0.684, 0.683, 0.682, 0.681$  for  $d = 4, \dots, 9$ . (3) For every  $d \geq 4$ ,  $v_c(C_d) < v_c(\text{DKZ}_d)$ : on  $\Phi_d$ , and on every  $\Phi_{kd}$ , the DKZ measurements do not realise the highest resistance of the violation of local realism to uniform outcome noise.*

**Lemma 9.3.** *Let  $s_{xy} = -1$  for  $x = y = 2$  and  $s_{xy} = 1$  otherwise,  $\text{CH}(a, b | x, y) = \frac{1}{4}(1 + (-1)^{a+b}s_{xy}/\sqrt{2})$  for  $a, b \in \{0, 1\}$  and 0 otherwise, and  $\delta_{00}(a, b | x, y) = [a = b = 0]$ . The behaviour of  $C_d$  is CH for even  $d$  and  $\frac{d-1}{d}\text{CH} + \frac{1}{d}\delta_{00}$  for odd  $d$ .*

*Proof.* The projections are real symmetric, so the behaviour is  $\text{Tr}(P_a^x Q_b^y)/d$ . For  $e, f \in \{\pm 1\}$ ,

$$\frac{1}{2} \text{Tr}\left(\frac{\mathbf{1} + eA_x}{2} \cdot \frac{\mathbf{1} + fB_y}{2}\right) = \frac{1}{4}\left(1 + \frac{ef}{2} \text{Tr}(A_x B_y)\right), \quad \frac{1}{2} \text{Tr}(A_x B_y) = s_{xy}/\sqrt{2}.$$

For odd  $d$  each of the  $(d-1)/2$  blocks contributes  $\text{Tr}(\text{IIII}')/d$ , and the last dimension contributes  $[a = b = 0]/d$ .  $\square$

Let  $\varsigma(0) = 1$ ,  $\varsigma(a) = -1$  for  $a \neq 0$ ,  $E_{xy}(p) = \sum_{a,b} \varsigma(a)\varsigma(b)p(a,b|x,y)$  and  $\mathcal{E} = E_{11} + E_{12} + E_{21} - E_{22}$  (CHSH after coarse-graining every measurement into  $\{0\}$  and the rest).

**Lemma 9.4.**  $\mathcal{E} \leq 2$  on  $\mathcal{L}_d$ ,  $\mathcal{E}(p_0) = 2(d-2)^2/d^2$ ,  $\mathcal{E}(\text{CH}) = 2\sqrt{2}$ ,  $\mathcal{E}(\delta_{00}) = 2$ . Hence the behaviour  $p_v$  of  $C_d$  violates  $\mathcal{E} \leq 2$  if and only if  $v > v_{\text{even}}(d)$  (even  $d$ ), respectively  $v > v_{\text{odd}}(d)$  (odd  $d$ ).

*Proof.* On  $\delta_\lambda$ ,  $\mathcal{E} = \varsigma(a_1)(\varsigma(b_1) + \varsigma(b_2)) + \varsigma(a_2)(\varsigma(b_1) - \varsigma(b_2)) = \pm 2$ . Under  $p_0$  the coarse-grained outcomes are independent with mean  $(2-d)/d$ . The threshold is  $(2 - \mathcal{E}(p_0))/(\mathcal{E}(p) - \mathcal{E}(p_0))$ , with  $2 - \mathcal{E}(p_0) = 8(d-1)/d^2$  and  $\mathcal{E}(p) - \mathcal{E}(p_0) = (2\sqrt{2}d^2 - 2(d-2)^2)/d^2$ , respectively  $2(d-1)((\sqrt{2}-1)d+4)/d^2$ .  $\square$

**Proposition 9.5** (a local model at the threshold). *Let  $d \geq 3$ ,  $s = 1/d$  and  $T = 4s(1-s)/(\sqrt{2} - (1-2s)^2) = v_{\text{even}}(d)$ . Then  $T\text{CH} + (1-T)p_0$  is local.*

*Proof.* Let  $c_1 = (1-T)(1-2s)$ ,  $c_2 = (1-T)(1-2s)^2$  and  $c_0 = 1 - 4c_1 + 3c_2 = 4s[(1-s) - (\sqrt{2}-1)(1-3s)]/(\sqrt{2} - (1-2s)^2) > 0$ ; also  $c_1 - c_2 = 2s(1-T)(1-2s) \geq 0$ . Let  $\Lambda \in \{0, 1, *\}^4$  (entries for  $A_1, A_2, B_1, B_2$ ) be  $(*, *, *, *)$  with probability  $c_2$ ; each of  $(*, r, r, 1-r)$ ,  $(r, *, r, r)$ ,  $(r, 1-r, *, r)$ ,  $(r, r, r, *)$ ,  $r \in \{0, 1\}$ , with probability  $\frac{1}{2}(c_1 - c_2)$ ; and each of the eight  $\lambda \in \{0, 1\}^4$  with  $\sum_{x,y} s_{xy}(-1)^{\lambda_{A_x} + \lambda_{B_y}} = 2$  with probability  $\frac{1}{8}c_0$ . Each party outputs its entry if it is 0 or 1, and an independent uniformly random element of  $\{2, \dots, d-1\}$  if it is  $*$ . Entries with  $a$  or  $b \geq 2$  come only from the first two families and equal  $(1-T)/d^2$ . For  $a, b \in \{0, 1\}$  the patterns of the second family with  $*$  elsewhere are aligned with  $s_{xy}$ , and six of the eight strategies of the third family are aligned on each pair of settings, so  $q(a, b|x, y) = \frac{1}{4}(1 - 2c_1 + c_2) + \frac{1}{8}(-1)^{a+b}s_{xy}(1 - c_2)$ . This equals  $T\text{CH}(a, b|x, y) + (1-T)d^{-2}$ , because  $1 - 2c_1 + c_2 = T + 4(1-T)s^2$  and  $1 - c_2 = \sqrt{2}T$ .  $\square$

*Proof of Theorem 9.2(1).* The upper bounds are Lemmas 9.3 and 9.4. Equality for even  $d$  is Proposition 9.5. For odd  $d$ ,  $vp_{C_d} + (1-v)p_0 = vs\delta_{00} + (1-vs)(v'\text{CH} + (1-v')p_0)$  with  $v' = v(1-s)/(1-vs)$ , and  $v' = T$  at  $v = v_{\text{odd}}(d)$ .  $\square$

**Lemma 9.6** (the  $\frac{1}{2}$ -lemma). *If all marginals of  $p$  are uniform, then  $p_v$  is local for  $v \leq \frac{1}{2}$ . The constant is optimal: the behaviour  $\text{PR}_d(a, b|x, y) = \frac{1}{d}[b - a \equiv [x=2][y=1]]$ , a  $d$ -outcome version of the box of Popescu and Rohrlich[26], has  $I_d(\text{PR}_d) = 4$  and  $v_c(\text{PR}_d) = \frac{1}{2}$ .*

*Proof.* The distributions

$$P_1(a_1, a_2, b_1, b_2) = p(a_1, b_1|1, 1)p(a_2, b_2|2, 2), \quad P_2(a_1, a_2, b_1, b_2) = p(a_1, b_2|1, 2)p(a_2, b_1|2, 1)$$

are local models; on the other two pairs of settings they give independent uniform outcomes. So  $\frac{1}{2}(P_1 + P_2)$  models  $\frac{1}{2}(p + p_0)$ , and  $p_v = 2v \cdot \frac{1}{2}(p + p_0) + (1-2v)p_0$ .  $\square$

The DKZ behaviour is  $p_{\text{DKZ}}(a, b|x, y) = \frac{1}{d}h_{xy}(b-a)$  with  $h_{xy}(m) = 1/(2d^2 \sin^2(\pi(m - \delta_{xy})/d))$ ,  $\delta_{xy} = \alpha_x + \beta_y$  (so  $\delta_{11} = \delta_{22} = \frac{1}{4}$ ,  $\delta_{12} = -\frac{1}{4}$ ,  $\delta_{21} = \frac{3}{4}$ ); its marginals are uniform, so  $v_c(\text{DKZ}_d) \geq \frac{1}{2}$ .

**Lemma 9.7** (covariant local models and absorption). *(i) If  $\mu$  is a probability distribution on  $M_d = \{m \in \mathbb{Z}_d^{\{1,2\}^2} : m_{11} - m_{12} - m_{21} + m_{22} \equiv 0\}$  with marginals  $\mu_{xy}$ , then  $\frac{1}{d}\mu_{xy}(b-a)$  is local. (ii) Let  $p$  and a local  $Q$  have uniform marginals and  $0 < v_0 < v_1 \leq 1$ . If  $|Q - p_{v_1}| \leq (v_1 - v_0)/(2d^2v_0)$  entrywise, then  $p_{v_0}$  is local.*

*Proof.* (i) With  $m \sim \mu$  and an independent uniform  $c$ , output  $A_1 = c$ ,  $A_2 = c + m_{11} - m_{21}$ ,  $B_1 = c + m_{11}$ ,  $B_2 = c + m_{12}$ ; then  $B_y - A_x = m_{xy}$ . (ii) With  $E = p_{v_1} - Q$  and  $\gamma = v_0/(v_1 - v_0)$ ,  $p_{v_0} = \frac{v_0}{v_1}Q + (1 - \frac{v_0}{v_1}) \cdot \frac{1}{2}(p_0 + S)$ , where  $S = p_0 + 2\gamma E \geq 0$  has uniform marginals, so Lemma 9.6 applies.  $\square$

**Proposition 9.8** (computer-assisted). *For  $3 \leq d \leq 20$  there are rationals  $v_0 < v_1$  and a distribution  $\mu$  on  $M_d$  with rational weights (denominator  $10^{16}$ ) such that  $|\mu_{xy}(m) - v_1 h_{xy}(m) - (1 - v_1)/d| \leq (v_1 - v_0)/(2dv_0)$ . Hence  $v_c(\text{DKZ}_d) \geq v_0(d)$ , with  $2/I_{\text{ME}}(d) - 1.02 \cdot 10^{-7} < v_0(d)$ ; in particular the bounds of Theorem 9.2(2) hold.*

*Proof.* The weights are checked in integer arithmetic. The values  $h_{xy}(m)$  are enclosed in rational intervals:  $\pi$  by partial sums of Machin's formula (width below  $10^{-50}$ ), and  $\sin$  by consecutive partial sums of its Taylor series on  $[0, 2]$ , after reduction to  $\sin(\pi r)$  with  $r \in [0, \frac{1}{2}]$ ; the condition is then decided exactly, with margins about  $10^{-13}$  against enclosure widths about  $10^{-15}$ . The distributions were found by a floating-point linear program[15] and rounded; this plays no role in the check. Program `dkz_cert.py` and the certificates are in `complete-resolution/noise-literal/`.  $\square$

*Proof of Theorem 9.2(2) and (3).* The upper bound follows from  $I_d(p_{\text{DKZ}}) = I_{\text{ME}}(d)$  and  $I_d(p_0) = 0$ . For  $d \geq 10$ ,  $v_c(C_d) < \frac{1}{2}$ : for even  $d$  this is  $(\sqrt{2} - 1)d^2 - 4d + 4 > 0$ , true for all  $d \geq 9$ ; for odd  $d$  it is  $d > 4(\sqrt{2} + 1) = 9.657\dots$ . For  $4 \leq d \leq 9$ , exact comparison in  $\mathbb{Q}(\sqrt{2})$  gives  $v_c(C_d) = 0.6442, 0.6589, 0.5729, 0.5798, 0.5137, 0.5176 < r(d)$ . On  $\Phi_{kd}$  use  $C_d \otimes \mathbf{1}_k$  and  $\text{DKZ}_d \otimes \mathbf{1}_k$ .  $\square$

**Proposition 9.9** ( $d = 2, 3$ ). (1) *For  $d = 2$  every quantum behaviour has  $v_c \geq 1/\sqrt{2} = v_c(\text{DKZ}_2)$  (positivity and the CHSH inequalities describe  $\mathcal{L}_2$ [10], CHSH vanishes on  $p_0$ , and Tsirelson's bound[5] holds).* (2) *(Computer-assisted.) Every projective strategy on  $\Phi_3$  has  $v_c \geq 2/I_{\text{ME}}(3) = 3\sqrt{3} - \frac{9}{2} = 0.696152\dots$ , with equality only for  $\text{DKZ}_3$  up to relabellings and  $u \otimes \bar{u}$ .* (3) *On  $\Phi_2$ , the measurements  $\{\Pi_+^x, \Pi_-^x, 0\}$ ,  $\{\Pi_+^y, \Pi_-^y, 0\}$  give  $v_c = 8/(9\sqrt{2} - 1) = 0.682132\dots < v_c(\text{DKZ}_3)$ ; the same holds on every  $\Phi_D$  with  $D$  even, in particular on  $\Phi_6$ , which also carries  $\text{DKZ}_3 \otimes \mathbf{1}_2$ .*

*Sketch of proof of (2) and (3).*  $\mathcal{L}_3$  has exactly 1116 facets: 36 positivity facets, 648 lifted CHSH facets and 432 relabelled CGLMP facets[6] (recomputed by an exact double description). The CGLMP facets are handled by Theorems 1 and 2 (relabellings map projective strategies on  $\Phi_3$  to such strategies and fix  $p_0$ ). For a lifted CHSH facet, Hölder's inequality and Jordan's lemma bound the quantum value by  $(4\sqrt{2} + 2)/3$ , while the noise contributes  $\pm \frac{2}{9}$ , so these facets hold for  $v \leq 4/(3\sqrt{2} + 1) = 0.762974\dots$ . Part (3) is Lemma 9.4 and Proposition 9.5 with  $d = 3$ .  $\square$

*Remark 9.10* (unused outcomes; scope).  $C_d$  leaves  $d - 2$  outcomes unused, while  $p_0$  gives each of them probability  $1/d$ . On  $\Phi_d$  a PVM whose  $d$  outcomes all occur is rank-one, and then  $p_0$  is the white noise of reading (c); so a projective counterexample on  $\Phi_d$  using every outcome would settle reading (c), which is open for  $d \geq 4$  (numerically, DKZ is optimal for  $d \leq 8$ : see ref. [12] for  $d \leq 5$  and the searches summarised in the repository; already for  $d = 4$  the polytope of link laws of covariant local behaviours, Masanes' polytope of correlation functions[21], has facets that are not CGLMP inequalities). With POVMs every outcome can be used: the effects  $(1 - \varepsilon)P_a^x + \frac{\varepsilon}{d}\mathbf{1}$ ,  $(1 - \varepsilon)Q_b^y + \frac{\varepsilon}{d}\mathbf{1}$  with  $(P, Q) = C_d$  and  $0 < \varepsilon \leq \frac{1}{100}$  are invertible and still beat DKZ for every  $d \geq 4$ ; and projective measurements on  $\Phi_D$  with  $D > d$  that use every outcome beat DKZ for  $(d, D) = (4, 12), (5, 11), (6, 12), (7, 13), (8, 12), (9, 13)$ .

*Prior work.* Acín, Durt, Gisin and Latorre[1] found the value  $v_{\text{even}}(d)$  (their eq. (14)) for the CHSH strategy on a Schmidt-rank-two maximally entangled state in  $\mathbb{C}^d \otimes \mathbb{C}^d$  with two-outcome measurements, and concluded that resistance to noise is not a good measure of nonlocality; Gruca, Laskowski and Żukowski[12] found numerically that Schmidt-rank-two states are the most noise-robust ( $d \leq 5$ ); Baek, Ryu and Lee[2] observed by linear programming that merged-outcome strategies beat DKZ on  $\Phi_4$  and  $\Phi_{16}$  under uniform noise (robustness 0.448 versus 0.552 for  $d = 4$ ), without relating this to Problem 27. The DKZ thresholds  $2/I_{\text{ME}}(d)$  were computed numerically in refs. [17, 8, 7, 12], and analytically for  $d = 3$  in ref. [4]. As far as we found, the exact theorem

for every  $d \geq 4$  on  $\Phi_d$ , the exact threshold of  $C_d$  in the  $d$ -outcome scenario, the rigorous lower bounds for  $\text{DKZ}_d$  and the consequence for Problem 27 are new.

**9.3. The Kullback–Leibler clause.** For a distribution  $\sigma$  on the setting pairs let  $U_q(\sigma) = \inf_{p \in \mathcal{L}_d} \sum_{x,y} \sigma_{xy} D(q(\cdot, \cdot | x, y) \| p(\cdot, \cdot | x, y))$  (in bits); the statistical strengths of van Dam, Gill and Grünwald[28] are  $S^{\text{UNI}}(q) = U_q(\frac{1}{4})$ ,  $S^{\text{UC}}(q)$  (supremum over product  $\sigma$ ) and  $S^{\text{COR}}(q)$  (supremum over all  $\sigma$ ). For  $d = 4k + \rho$  ( $k \geq 1$ ,  $\rho \in \{0, 1, 2, 3\}$ ) let  $B_d$  be the strategy on  $\Phi_d$  that uses, on consecutive coordinate blocks with disjoint outcome groups,  $k$  copies of two CHSH strategies with shared settings (product bases of the eigenvectors of  $A_x$ , resp.  $B_y$ , outcome  $2a_1 + a_2$ ) and one block of size  $\rho$  (a single basis vector, the CHSH bases, or the DKZ bases for  $d = 3$ ). Its four measurements are orthonormal bases of  $\mathbb{C}^d$ .

**Theorem 9.11** (computer-assisted). *For every  $d \geq 4$  and  $X \in \{\text{UNI}, \text{UC}, \text{COR}\}$ ,*

$$S^X(\text{DKZ}_d) \leq 0.0687802606 < 0.0703203697 \leq S^{\text{UNI}}(B_d) \leq S^X(B_d),$$

*and  $S^{\text{UNI}}(B_d) \geq 0.0879004622$  if  $4 \mid d$ . The same holds on  $\Phi_{kd}$  for  $\text{DKZ}_d \otimes \mathbf{1}_k$  and  $B_d \otimes \mathbf{1}_k$ .*

*Sketch of proof. Test factors.* If  $r \geq 0$  is positive on the support of  $q$  and  $\sum_{x,y} \sigma_{xy} r(a_x, b_y | x, y) \leq 1$  for every  $\lambda$ , Jensen’s inequality gives  $U_q(\sigma) \geq \sum_{x,y} \sigma_{xy} \sum_{a,b} q \log r$ . *Competitor.* The behaviour of  $B_d$  is the mixture  $\sum_j \frac{m_j}{d} q_j$  of its block behaviours, supported on disjoint outcome sets. For uniform  $\sigma$ , block test factors with entries at most 2 add: a deterministic  $\lambda$  either has all its contributing pairs of settings in one block, or exactly two diagonal pairs in two blocks, contributing at most  $\frac{1}{4}(2 + 2) = 1$ . Exact rational test factors (entries at most 1.24801, feasibility checked by exhaustive enumeration) certify the block values 0, 0.046273845411, 0.057783024051 and  $L_4 = 0.087900462237$  bits; the resulting bound  $(4kL_4 + \rho L_\rho)/d$  is nondecreasing in  $k$  and at least  $\frac{4}{5}L_4 \geq 0.0703203697$ . *DKZ.* By the partial fraction expansion of  $\pi^2/\sin^2(\pi z)$ ,  $h_{xy}$  is the reduction modulo  $d$  of the distribution  $h_{xy}^\infty(m) = 1/(2\pi^2(m - \delta_{xy})^2)$  on  $\mathbb{Z}$ . A distribution  $\mu$  of  $(t, n_1, n_2) \in \mathbb{Z}^3$  defines, with a uniform shift  $c$ , the local model  $A_1 = c$ ,  $A_2 = c + t$ ,  $B_1 = c + n_1$ ,  $B_2 = c + n_2$  modulo  $d$ , and data processing gives  $S^{\text{UNI}}(\text{DKZ}_d) \leq T(\mu) = \frac{1}{4} \sum_{x,y} D(h_{xy}^\infty \| \mu_{xy})$  for every  $d$ , where  $\mu_{xy}$  are the laws of  $n_1, n_2, n_1 - t, n_2 - t$ . An explicit rational  $\mu^*$  with tail  $\propto 1/(t^2 - \frac{1}{4})$  gives  $T(\mu^*) \leq 0.068780260568$  (interval arithmetic for  $|m| \leq 3000$ ; nonpositive terms beyond). Two relabellings fix  $p_{\text{DKZ}}$  and generate a group that acts transitively on the setting pairs, so by concavity of  $\sigma \mapsto U_q(\sigma)$ ,  $S^{\text{UNI}} = S^{\text{UC}} = S^{\text{COR}}$  for  $\text{DKZ}_d$ .  $\square$

Moreover  $S(\text{DKZ}_d) \leq S(\text{DKZ}_{kd})$ , and  $0.0687464 \leq \sup_d S(\text{DKZ}_d) \leq 0.0687711$ . For  $d = 3$ ,  $\text{DKZ}_3$  is a strict local maximiser of all three strengths among projective strategies on  $\Phi_3$ , modulo phases and local unitaries  $u \otimes \bar{u}$  (computer-assisted: exact Karush–Kuhn–Tucker data in  $\mathbb{Q}(\sqrt{3})$ , with an optimal test factor that is affine in the CGLMP level, and an interval Cholesky certificate for the averaged Hessian), with  $S(\text{DKZ}_3) = 0.05778302549332865$  bits; global optimality at  $d = 3$  is supported numerically but open. The clause was first shown to fail for  $d = 4$  by Zhang[29], with an exact certificate, and for  $4 \leq d \leq 9$  in our earlier work[24]. Programs and certificates (`verify_main.py`, `zcert.py`, `blocks_cert.py`, `d3_cert.py`) are in `complete-resolution/kl-divergence/`.

## 10. FORMAL VERIFICATION IN LEAN 4

The argument of this Supplementary Information has been formalised in the Lean 4 theorem prover[19] with the mathematical library Mathlib[22], in the versions

Lean 4.33.1,    Mathlib revision 0df444a360eaa60ab8c11dca51a86af692955474.

The formalisation has 96 files and about 33,000 lines; it is in the directory `lean/` of the repository[25]. No file uses `sorry`, `admit`, an axiom declaration or `native_decide`, and

#print axioms reports, for every main theorem, only the three standard axioms `propext`, `Classical.choice` and `Quot.sound`. The main theorems (file `Main.lean` and the files it imports, and the file `CgImpNoise.lean`, which imports `Main.lean`) are the following.

| Lean theorem                                                                                                     | statement                                                                                                                                             | assumptions                                                                    |
|------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|
| <code>QQP27.maxEntClause_le_twenty</code>                                                                        | for every $2 \leq d \leq 20$ : Theorems 1 and 2, and the DKZ strategy attains $I_{\text{ME}}(d)$                                                      | none                                                                           |
| <code>QQP27.maxEntClause_all</code>                                                                              | Theorems 1 and 2 for every $d \geq 2$                                                                                                                 | $\text{CONE}_d$ for $d \geq 21$ only<br>( <code>Hyp_ConeCertPos_large</code> ) |
| <code>QQP27.stripInequality</code> ,<br><code>QQP27.stripEquality</code>                                         | Theorem 3 for every matrix size $M$ , and its equality case                                                                                           | none                                                                           |
| <code>QQP27.StripL3b.theorem1_bmv2</code>                                                                        | Theorem 4 (two-variable BMV formula with explicit density), for all $(a, t) \in \mathbb{C}^2$                                                         | none                                                                           |
| <code>QQP27.StripL3b.hyp_RI</code>                                                                               | the Radon identity (Proposition 3.11)                                                                                                                 | none                                                                           |
| <code>QQP27.Red.hyp_reduction</code>                                                                             | the exact reduction: strategy $\rightarrow$ clock model $\rightarrow$ 27B configuration $\rightarrow$ linear form (Section 2)                         | none                                                                           |
| <code>QQP27.Cell.continuumCell</code>                                                                            | the continuum theorem for step fields: Theorem 3 implies every cell inequality (Sections 5, 6)                                                        | none                                                                           |
| <code>QQP27.ClassB.classicalTheoremB_all</code>                                                                  | the classical (commuting) theorem for every $d$ , including the discrete Hilbert-transform rearrangement ( $N = 4d$ , $ A  =  B  = d$ )               | none                                                                           |
| <code>QQP27.coneCertPos_le_twenty</code>                                                                         | $\text{CONE}_d$ for $2 \leq d \leq 20$ : the explicit certificates of Section 7 re-evaluated by the Lean kernel in exact rational interval arithmetic | none                                                                           |
| <code>QQP27.ConeCertificate.fullCheck_sound</code>                                                               | soundness of the certificate checker                                                                                                                  | none                                                                           |
| <code>QQP27.dkz_cgImp</code>                                                                                     | the DKZ strategy attains $I_{\text{ME}}(d)$ , for every $d$                                                                                           | none                                                                           |
| <code>QQP27.cgImpOf_noisy</code> ,<br><code>QQP27.noisy_violation_iff</code><br>( <code>CgImpNoise.lean</code> ) | uniform noise of weight $1 - v$ multiplies $I_d$ by $v$ ; the CGLMP noise threshold of a strategy is $2/I_d$                                          | none                                                                           |
| <code>QQP27.noisy_violation_imp_le_twenty</code> , <code>QQP27.noisy_threshold_rigid_le_twenty</code>            | Proposition 9.1 for $2 \leq d \leq 20$                                                                                                                | none                                                                           |
| <code>QQP27.noisy_violation_imp</code> ,<br><code>QQP27.noisy_threshold_rigid</code>                             | Proposition 9.1 for every $d \geq 2$                                                                                                                  | $\text{CONE}_d$ for $d \geq 21$ only                                           |

Hence the maximally entangled clause of Open Quantum Problem 27B is fully machine-checked for  $2 \leq d \leq 20$ . For  $d \geq 21$  the formal proof is complete except for the single input  $\text{CONE}_d$ , stated in the finite form of Remark 7.3, which is established by the interval-arithmetic certificates of Section 7, outside Lean. The formal proof of Theorems 3 and 4 follows the elementary route of Section 3: the Radon identity, then Theorem 4 by Tonelli's theorem and the identity theorem; its one-variable uniqueness step uses the Fourier inversion theorem of Mathlib. The continuum theorem is formalised for step fields, with explicit Herglotz functions, as in Section 5.

Theorems 3 and 4 are formalised in full in the following sense: the identity (3.1) on  $\mathbb{C}^2$ , with  $F \geq 0$  measurable, compactly supported and bounded by  $C(\tau(1 - \tau))^{-1/2}$ , and the strip inequality with its exact defect formula and its equality case, for every matrix size. A few auxiliary statements are proved on paper only, because the formal proof does not need them: the continuity

of  $F$  (Lean uses measurability, and a bound in Frobenius norm), the sum rule of Remark 3.6, and the dilogarithm formula for  $h_\lambda$  (Lean defines  $h_\lambda$  as the Poisson integral of Lemma 4.2). Likewise, the formal rigidity statement is the main assertion of Theorem 2 ( $d$  divides  $D$ , and a unitary  $u$  maps the measurements to  $\text{DKZ} \otimes \mathbf{1}$ ); the uniqueness of  $u$  and the corollary for  $d \nmid D$  (Step 4 of Section 8) are proved on paper.

## 11. INDEPENDENT VERIFICATION AND CHECKS

*Independent verification.* Each proof was refereed by an independent checker, who re-derived it step by step and wrote the numerical checks from scratch; the reports are in the repository[25].

- (1) Theorems 3 and 4, by the route of Sections 3 and 4 (Paley–Wiener–Schwartz and Fourier slices): verdict correct. Equation (3.1) was confirmed at 82 complex points  $(a, t)$  to  $6.8 \times 10^{-13}$ , and the defect formula of Theorem 4.3 at 51 pairs (instance,  $\lambda$ ) to  $1.3 \times 10^{-12}$ .
- (2) The Radon identity and the elementary route (Proposition 3.11 and its consequences): verdict correct; the identities were confirmed to 34–46 digits.
- (3) The reduction chain, the continuum theorem, the cell inequalities, the logic of  $\text{CONE}_d$  and the rigidity argument (Sections 2, 5–8): verdict correct.
- (4) The alternative proof of Theorem 3 for  $\text{rank } B \leq 1$  (and hence for  $\text{corank } B \leq 3$  and all  $M \leq 5$ ): verdict correct (no mathematical error; some expository gaps, none of which changes a statement).
- (5) Theorem 9.2, two independent verifications: verdict correct, no error and no gap. The first re-derived every step, recomputed the facets of  $\mathcal{L}_3$  independently, re-verified all eighteen certificates of Proposition 9.8 in 200-bit interval arithmetic and ran eleven tamper tests on the verification program, all of which were detected; the second rebuilt the result from scratch, with its own certificates, and found the local model of Proposition 9.5.
- (6) Theorem 9.11 and the statements on  $d = 3$  and on  $\sup_d S(\text{DKZ}_d)$ : verdict correct, no error; the bound  $T(\mu^*) \leq 0.068780260568$  was reproduced digit for digit and the local optimality at  $d = 3$  re-proved with a different chart and slice. The sharper bounds on  $\sup_d S(\text{DKZ}_d)$  come from this verification.

*Checks.* The following checks are not part of the proofs; they guard against errors in them. All scripts and logs are in the repository[25].

| item                                    | check                                                                                                                                                                                                                                                                                                                                                                                 |
|-----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Theorem 3.1                             | independent re-implementation: equation (3.1) to $6.8 \times 10^{-13}$ (relative) at 82 complex $(a, t)$ on 13 instances ( $M \leq 6$ ); back-projection identity (step 2) to $10^{-20}$ at 15 random $(s, \tau)$ with 0, 2 and 4 non-real roots; sum rule (Remark 3.6) to $10^{-13}$ ; Radon identity, which gives the Radon slices, to 34–46 digits on 22 instances with $M \leq 8$ |
| Theorem 4.3                             | defect formula to $1.3 \times 10^{-12}$ at 51 (instance, $\lambda$ ) pairs on 9 instances, including spectra up to $\pm 150$ , clustered spectra and $M = 6$ (independent code); adversarial searches for violations of the strip inequality for $M \leq 8$ (none); second proof for rank $\leq 1$ , corank $\leq 3$ , $M \leq 5$                                                     |
| Lemma 2.2, Prop. 6.1                    | identities checked on random non-commuting configurations for $3 \leq d \leq 12$ , $M \leq 4$ , to $10^{-13}$ ; every rotated cell inequality negative on random configurations                                                                                                                                                                                                       |
| Theorem 5.1                             | chain (5.1) term by term on step fields ( $d = 4$ , $M = 2, 3$ ); mean-value identity; $\lambda^*$ and $\Phi_c(\frac{1}{4}, \frac{1}{4}) = G/\pi^2$ to 40 digits                                                                                                                                                                                                                      |
| Theorem 7.1                             | separate checker and complete re-verification for $d \leq 200$ , and re-evaluation by the Lean kernel for $d \leq 20$ ; floating-point re-evaluation; audit of all 1800 single-run certificates for $201 \leq d \leq 2000$ ; non-rigorous prototypes for large $d$ ; audit of box coverage and margins for $d \geq 2001$                                                              |
| Rigidity                                | residue formula and Herglotz boundary values against quadrature; gradient ascent: all runs that reach the maximum end at commuting families                                                                                                                                                                                                                                           |
| Theorems 1 and 2,<br>$3 \leq d \leq 20$ | independent exact sum-of-squares certificates[24]                                                                                                                                                                                                                                                                                                                                     |

## 12. DETAILS OF THE NUMERICAL ILLUSTRATIONS

All figures are produced by `make_figures.py`. Fig. 1b: Riemannian gradient ascent with backtracking on  $U(D)^4$  (the four measurement bases, rank-one projectors for  $D = 5$  and rank-two projectors for  $D = 10$ ), 20 runs each from Haar-random starting points; the ascent direction for a basis  $U$  with blocks  $\Pi_a$  and gradient operators  $G_a$  is  $\sum_a [X_a, \Pi_a]$ ,  $X_a = U^\dagger G_a U$ . Figs. 3 and 4: a random instance ( $M = 5$ , rank  $B = 2$ , fixed seed);  $h_\lambda$  is evaluated by the dilogarithm with principal branch; the breakpoints of  $F(\cdot, \tau)$  are the critical values of the eigenvalue curves  $x \mapsto \lambda_j(g - x(P - \tau))$ , located by bisection; the integrals are computed by adaptive Gauss–Kronrod quadrature after the substitutions  $s = s_a + (s_b - s_a) \sin^2(\theta/2)$  on each interval between breakpoints and  $\tau = \sin^2(\vartheta/2)$ . The defect of the strip inequality and the balayage of  $F$  agree to  $1.4 \times 10^{-14}$ , and the  $\tau$ -marginals equal  $\|BgP\|_F^2$  to  $6 \times 10^{-14}$ . Fig. 5 reads the certificate files directly: the LP certificates for  $d \leq 200$ , the single-run certificates for  $201 \leq d \leq 2000$  (the plotted values are the lower endpoints of the certified enclosures), and the box logs of the final run for  $d \geq 2001$ .

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